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Improving the Circularization Efficiency, Stability and Translatability of Circular RNA by circDesign

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Abstract

Synthetic circular RNA (circRNA) has recently emerged as the next generation RNA technology for vaccine and drug development, featuring its uniqueness in a closed-loop structure and cap-independent translation mechanism. However, rational design and optimization of circRNA to improve its stability and protein coding potential remains challenging due to lack of structure prediction tool and sequence design algorithm for endless circular-shaped RNA. In this study, we presented circDesign, a computational approach to achieve optimal design of the circRNA with enhanced circularization efficiency, stability, and translatability. Using rabies virus glycoprotein (RABV-G) and varicella-zoster virus (VZV) glycoprotein gE as the exemplified antigens, we demonstrated that optimized circRNAs encoding corresponding antigens exhibited higher stability and protein translation efficiency *in vitro*. Formulated by proprietary delivery system lipopolyplex (LPP), the optimized circRNA vaccines encoding RABV-G or VZV gE antigen further elicited enhanced humoral and cellular *in vivo* immune responses in mice. Taken together, this circular RNA design approach provided a general strategy to leverage the capability of circRNA as next generation vaccines or therapeutics.

Keywords: circular RNA; circDesign; Algorithm; RNA vaccine; Rabies vaccine; Varicella-zoster virus vaccine; Lipopolyplex

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Introduction

Rational sequence design of messenger RNAs (mRNAs) is of pivotal importance to maximize the potency of mRNA vaccines or therapeutics[1]. Optimization of conventional linear mRNA usually involves 5'- & 3'-untranslated region (UTR) and open reading frame (ORF), which could be simply achieved by combining naturally occurring UTR pair with codon optimized ORF region[2, 3]. As exemplified by Moderna's mRNA-1273 and BioNTech's BNT162b2 COVID-19 mRNA vaccines, the optimized ORF was directly flanked with previously screened UTR pair[4]. Besides, it was reported the functional half-life of mRNA dictates its *in vivo* performance as the translation efficiency and stability of mRNA synergistically impact the protein output[5, 6]. To achieve enduring and efficient translation, the ORF sequence should be optimized towards more stable secondary structure without compromising its codon preference.

Recent advancement in circRNA demonstrated its potentials as next generation vaccines or therapeutics owing to its uniqueness in endless structure distinct from linear mRNA[7-9]. Other than ORF region, synthetic circRNA usually contains internal ribosome entry site (IRES) motif upstream of ORF for translation initiation[10]. The correct folding of IRES as a functional motif to recruit translation pre-initiation complex (PIC) is a prerequisite of efficient translation of circRNA[11]. However, due to the lack of structure prediction tool for circular RNA, along with insufficient understanding of the crosstalk between IRES and ORF regions on circRNA, current sequence optimization of circRNA heavily relied on modular screening[12]. This independent ORF optimization on circRNA without considering the backbone sequence including IRES motif and residual regions as a whole introduces more uncertainties in the process of circRNA sequence optimization.

Here we present an efficient approach called circDesign based on previously reported LinearDesign algorithm[6] to generate optimized circRNA sequences for rabies and VZV vaccine development. In brief, a pool of candidate ORFs were firstly generated using LinearDesign algorithm based on optimization of codon usage and minimum free energy (MFE). By assembly of ORF with IRES motif as well as other functional regions, the positional entropy and ensemble diversity were evaluated aside from overall MFE and codon adaptation index (CAI)[13-15]. To

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evaluate the impact of these parameters on circRNA performance, we performed *in vitro* and *in vivo* characterization of these circRNAs compared to conventionally optimized ORF sequences, it demonstrated greater performance in terms of circularization, stability, protein translation as well as *in vivo* immune responses. The circDesign approach provides a general strategy for protein-coding circular RNA design and potentially enables broader circRNA-based vaccine and therapeutic development.

Results

circDesign approach for circular RNA design

To implement circDesign for circRNA-based rabies vaccine design, the ORF region encoding RABV-G[16, 17] was designed such as to achieve efficient and prolonged expression. Different from linear mRNA design which is composed of relatively short functional elements (~ 100 nt in length) including 5'-UTR and 3'-UTR, artificially constructed circular RNA with protein translation capacity usually contains long viral or cellular IRES sequence (500-800 nt in length) for cap-independent translation[10, 18, 19]. More importantly, the IRES region needs to be independently folded into a hierarchical structure without interference from other sequence as to efficiently recruit translational PIC. Therefore, a key consideration in circDesign involves mitigating the cross-interaction of ORF region with IRES motif, thus facilitating the formation of functional IRES with high confidence. Moreover, following a permuted intron-exon (PIE) circularization[20], residual exon fragments on circularized RNA is redundant and introduced more uncertainties to the function of circRNA as this element may interfere with the folding of other elements. As such, precautions must be taken when optimizing the ORF of circRNA since functional IRES motif and residual exons are present in the sequence.

In this study, the circDesign approach consisted of four integral steps as illustrated in **Fig. 1a**. In brief, the first step involves the implementation of LinearDesign algorithm to generate a pool of ORF candidates. Followed by the screening of ORF based on CAI and MFE, the stability and translation efficiency were carefully considered. Then a completed circular RNA was assembled with other necessary residual fragments including IRES, exons and spacers. In addition, the positional entropy of IRES and ensemble diversity of circRNA were evaluated. Lower positional entropy suggests a more conserved structure of IRES, while a lower ensemble diversity indicates a higher consistency and predictability of the circRNA structure, both of which contribute to higher structure stringency[21, 22]. In addition, the inter-region and intra-region interaction was evaluated primarily aiming to mitigate the impact from ORF on the independent folding of IRES, which is critical for efficient translation initiation. Based on the

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abovementioned screening process, the designed circRNAs were then subjected to *in vitro* and *in vivo* evaluation for screening.

Design, synthesis, and characterization of circular RNA encoding RABV-G

Herewith RABV-G encoding circRNA, we designed six ORFs, along with two benchmark sequences. The six circDesign-generated designs are widely dispersed in the low-MFE regions (**Fig. 1b**). To better understand the impact of MFE and CAI in terms of biological implications, following the LinearDesign's experimental design principle, we crafted sequences with nearly identical values in MFE (CR2, CR3 // CR4, CR5) or in CAI (CR5, CR6, CR-Ther // CR1, CR2, CR-Novo). Benchmark ORF sequences (CR-Ther and CR-Novo) were designed with publicly available platforms from Thermo Fisher and Novoprotein respectively, which primarily optimized the codon preference without considering the secondary structure. The color darkness of each sequence represents the ensemble diversity of assembled circular RNA. As suggested, the ensemble diversity of designed circRNA is correlated with MFE. With a lower MFE of ORF region, the ensemble diversity of circRNA was lowered, suggesting a more well-defined structure with independent folding of each region.

To synthesize the circRNA, the circularization of RNA was mediated by self-cleaving permuted group I catalytic intron which was separated into 5'-intron and 3'-intron in the original sequence design as previously reported[7, 20]. In brief, both ends of linear mRNA consisted of complementary homology arms facilitating the folding of active ribozyme structure, thus undergoing two constitutive transesterification reactions to form junction between two exon fragments (E1 & E2). In this study, IRES from Coxsackievirus B3 (CVB3)[23, 24] was adopted upstream of open reading frame (ORF) to drive the protein translation process. As shown in the **Fig. 1b**, different ORF embedded in the linear mRNA design could be incorporated in circular RNA *via* the same circularization reaction. Taking CR3 and CR-Novo as representative sequences, the ORF of CR3 was designed with much lower MFE and less inter-region interaction with IRES motif compared to CR-Novo. As shown in **Fig. 1c**, though with the same backbone sequence and amino acid sequence, most of the amino acid could be coded by different synonymous codons, resulting in numerous sequence design space. The secondary structure of CR3 and CR-Novo was predicted as **Fig. 1d**. It was clear that the same IRES region (light grey color) had distinct folding in two circRNAs. The secondary structure folding prediction of circRNA rabies sequences (see **Extended Data. 1&2**) demonstrated that suboptimal ORFs (Ther and Novo) may disrupt the correct folding of IRES. And the pairwise visualization of these two circRNA (**Fig. 1e**) showed that CR-Novo had more inter-region base pairing (red connecting line) than that of CR3. The base pairing in CR3 mainly led to intra-region folding (blue connecting line).

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Same results were obtained in other circDesign-based sequences (see **Extended Data. 3&4**).

As a result of the difference in MFE, CAI value and ensemble diversity, the circularization efficiency varied among these sequences. Considering a more compact ORF region folding with less interference with the formation of active intron, circDesign-generated sequences (CR1-6) had relatively higher circularization efficiencies than those of CR-Novo and CR-Ther (**Fig. 2a**). To remove impurities from the circularization reaction products, size-exclusion column chromatography was used to obtain purified circRNA (see **Extended Data. 5a**) which induced minimized innate immunogenicity (see **Extended Data. 5b-c**). To characterize the circRNA, non-denaturing agarose gel electrophoresis was performed at room temperature. As shown in **Fig. 2b**, it's intriguing to find that the gel mobility of circRNAs was well correlated with their MFE values given their same nucleotide length. CR-Novo sequence, with the highest MFE, exhibited the slowest mobility in the gel matrix possibly due to its less compact structure, while circDesign-generated sequences especially CR1-5 had faster migration as a result of their highly compact folding structure. This suggested the potential of optimizing ORF sequences of circRNA in regulating their folding paradigm.

Optimized circRNA exhibited enhanced *in vitro* stability

As the secondary structure dictates the functionally half-life of mRNA[5], we then extensively evaluated the *in vitro* stability of circRNAs in terms of thermodynamic folding, in-solution degradation as well as intracellular decay. By heating the circRNA in solution to gradually unfold its structure, the fluorescence intensity from RNA bound RiboGreen® reagent attenuated accordingly. We calculated the melting temperature (T_m) of circRNA from the fluorescent intensity curve as **Fig. 2c**. CR1-3 had slightly higher T_m than CR4-6, with an obviously increased thermodynamic stability compared to CR-Novo and CR-Ther. The melting curve results was associated with MFE values.

Like linear mRNA, the chemical instability of circRNA originated mainly from hydrolysis reaction despite of its endless structure[25-27]. To examine the in-solution stability of all circRNAs, RNA samples were incubated in PBS buffer containing 10 mM Mg^{2+} at 37 °C mimicking the physiological buffering system over a time course of 60 h. At different time points, the integrity of circRNA was assessed. The degradation of circRNA as a function of time was fitted to a one-phase decay curve, suggesting different degradation kinetics. circRNAs with higher MFE values (CR-Novo & CR-Ther) underwent faster degradation kinetics with half-life ($t_{1/2}$) of 9.14h and 9.68h respectively, while CR1-3 had considerably extended $t_{1/2}$ ranging from 15.56 to 20.62 h (**Fig. 2d**).

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Apart from chemical degradation, the intracellular decay of exogenous circRNA is also regulated by different pathways in related to its translational activity[28, 29]. To assess the in-cell half-life of circRNA, the abundance of circRNA extracted from cells was quantified using reverse transcription quantitative real-time PCR (RT-qPCR) method at various time points (3, 12, 24, 48 and 72h) post-transfection into Human Skeletal Muscle Cells (HSkMC). As illustrated in **Fig 2e**, a slightly different decay pattern of all tested circRNA was observed. CR1 and CR3 had a prolonged intracellular half-life (24.34 h and 19.31 h, respectively) while CR-Novo and CR5 showed a faster in-cell decay (10.21 h and 9.18 h, respectively). As the biological fate of transfected circRNA is associated with a plethora of factors including translation status and nuclease resistance[30, 31], the in-cell degradation of artificially synthesized circRNA could possibly be controlled by tuning its secondary structure and translation efficiency. We reasoned that CR3 sequence had decent stability in conjunction with high translation efficiency, enabling its resistance against chemical degradation and biological decay.

circDesign improved the protein coding capacity of circRNA

Since the stability and translation efficiency of circRNA co-regulates the protein output level, we evaluated the RABV-G expression from transfected circRNA in HSkMC using flow cytometry (**Fig. 2f**). To gain thorough comparison, we included conventional chemically modified linear mRNA harboring Novo ORF encoding RABV-G as a control group as well. At 24 h post-transfection, antigen was highly expressed in CR3 group as suggested by the mean fluorescent intensity (MFI). CR1-5 had a remarkably higher protein output level than CR6 and CR-Ther, largely attributed to the relatively instable structure of CR6 and CR-Ther. However, it's interesting to find that benchmark CR-Novo exhibited a decent protein coding capacity despite of its lower CAI value compared to CR-Ther. This evidence emphasized the importance of taking multiple influential factors into consideration other than CAI to stratify the mRNA coding potentials. Moreover, we found that the circular form (CR-Novo) outcompeted its linear counterpart (LR-Novo) in protein translation level (**Fig. 2f**), where stability factor might be the major contributing factor as these two forms of RNA shared the same CAI of ORF. The correlation analysis of in-solution and in-cell half-life with MFE (**Fig. 2g**) showed the importance of MFE parameter in improving circRNA half-life. Predicted half-life correlated well with the observed experimental data (**Fig. 2h**).

To elucidate the underlying mechanism of enhanced protein expression in relation to structure (**Fig. 3a&b**), we performed ribosome profiling[32] and polysome profiling[33] analysis of CR3 and CR-Novo for side-by-side comparison. Following the standard workflow (**Fig. 3c**) after transfection into cells, the ribosome

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footprints were aligned to the full-length CR3 and CR-Novo, respectively. The ultracentrifugation fractions were quantified to determine the fractional composition of circRNA. In both RNA direct sequencing counts and ribosome sequencing counts (**Fig. 3d**), the CR3 had higher fpkm (fragment per kilobase of transcript per million mapped reads) of RNA-seq and ribo-seq than that of CR-Novo at the time of cell being harvested, indicating a higher circRNA abundance and ribosome footprints of CR3. Footprint abundance relative to mRNA abundance dictates the TE (translation efficiency) [34] of circular RNA. The TE of CR3 (233.7) was calculated as 1.5-fold of CR-Novo (159.9), suggesting a markedly higher translation efficiency from circDesign optimized CR3. After alignment of the ribosome footprints on circRNA, it's clear that the cap-independent translation mechanism of circRNA driven by CVB3 IRES efficiently recruits ribosome onto circular RNA to initiate the protein translation. Most of the ribosome protected fragments were identified in ORF region in both CR3 and CR-Novo, though the read count and distribution varied between these two groups. As suggested in **Fig. 3e**, the ribosomal footprints were mapped and distributed more evenly throughout the ORF region while CR-Novo had more distinct footprint accumulation peaks. The fully aligned ribosomal footprints were plotted on circular CR3 and CR-Novo annotated with each functional regions, respectively (**Extended Data. 6**). In addition, several major peaks upstream of ORF indicated the recruitment of ribosome for translation initiation. Taking a closer look at the secondary structure and pairwise visualization (**Extended Data. 7**), it's highly possible that the disrupted IRES motif in CR-Novo impeded the translational machinery recruitment, thus resulting in a much lower ribosome footprint abundance.

In addition, polysome profiling was performed on CR3 and CR-Novo to compare the ribosome loading capacity. Ultracentrifugation of both CR3 and CR-Novo was successful as shown in **Fig. 3f**, which quantified the total RNA in each fractional position. 80S fraction referred to monosome and later peaks referred to polysome-bound circRNA. By further quantifying the CR3 or CR-Novo in each collected fractions, we observed a higher ribosome loading in CR3 compared to CR-Novo (**Fig. 3g**).

LPP delivery system efficiently delivered circRNA *in vitro* and *in vivo*

To investigate the performance of circRNA *in vivo*, a proprietary lipopolyplex (LPP) delivery system[35, 36] was used to formulate the circular RNA. As shown in **Fig. 4a**, following a two-step microfluidic mixing process, cationic polymer complexed circRNA could be encapsulated in a bilayer liposome-like nanoparticle with a "core-shell" structure. Previous studies have demonstrated the efficiency and safety of using this LPP delivery system to deliver linear mRNA[37, 38]. Herewith circRNA encoding GFP (CR-GFP) or firefly luciferase (CR-FLuc) was formulated into LPP to study its *in vitro* cell expression and *in vivo*

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biodistribution compared to its chemically modified linear counterpart (LR-GFP or LR-FLuc), respectively. Flow cytometry analysis of cells treated with LPP formulated CR-GFP showed a dosage-dependent translation from 50 ng, 200 ng to 500 ng per well in a 6-well plate. Stronger expression was observed in circular RNA group than that of LR-GFP (**Fig. 4b**). Besides, the protection of encapsulated circRNA against RNase digestion was evaluated in LPP and lipid nanoparticle (LNP), a widely used mRNA delivery system. The residual circRNA was quantified at 30 min post-treatment and the results suggested a higher protection efficacy against nuclease digestion using LPP formulation.

Upon intramuscular administration of LPP formulated CR-FLuc and LR-FLuc in mice, *in vivo* imaging system (IVIS) captured the whole-body *in vivo* luminescence activities at different time points (6, 24, 48, 72, 96 and 168 h) (**Fig. 4d**), reflecting the expression kinetics of luciferase from CR-FLuc or LR-FLuc. It turned out that circular RNA produced luciferase in a more durable manner with higher accumulated expression as shown in **Fig. 4e**. Though with similar physicochemical properties of LPP formulation (**Extended Data. 8a**) and lower onset expression as seen at 6 h post administration, CR-FLuc showed extended expression over the course of one week while luminescence from LR-FLuc group descended to background level within a relatively shorter time (2 days post-dosing). A gross estimation of accumulated expression was conducted simply by calculating the area-under-curve (AUC) of total flux of luminescence. CR-FLuc group led to approximately 5.3-fold of total luciferase expression compared to its linear counterpart over the course of 8 days (**Fig. 4f**). Indeed, other group has reported similar translation kinetics of circular RNA compared to linear mRNA[12], thus rendering circRNA a promising therapeutic or vaccine platform for sustained protein supplement or enduring protection against pathogens. These results suggested the feasibility of using LPP for efficient *in vivo* delivery of circRNA despite of its closed loop feature.

circDesign optimized circRNA rabies vaccine demonstrated superior *in vitro* stability and *in vivo* performance

It has been reported that linear mRNA encapsulated in lipid nanoparticles still underwent slow transesterification hydrolysis[25]. In this study, the stability of LPP-circRNA nanoparticles was evaluated to simulate the vaccine storage condition under the formulated situation. Henceforth, the same lipid composition was adopted to encapsulate circDesign-optimized CR1-6 and benchmark circRNA encoding RABV-G to rule out the impact from delivery system on their *in vitro* and *in vivo* performance. The *in vitro* characterization of formulated circRNAs showed that LPP nanoparticles had similar size of around 110 nm (**Fig. 4g**) and positive zeta potentials (in the range of 20-35 mV) (**Fig. 4h**) regardless of the encapsulated circRNA. However, the importance of optimizing the stability of

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circRNAs was manifested in their storage stability in lipid formulation. We examined the mRNA content of LPP-circRNA complex on day 30 at room temperature to assess the integrity of RNA component. A faster RNA degradation in LPP formulated CR-Novo and CR-Ther was observed (**Fig. 4i**) with 15-20% circRNA loss on day 30 while circDesign optimized circRNAs (CR1-5) showed little degradation over tested period. Moreover, the in vitro innate immunogenicity and cell cytotoxicity was evaluated with LPP formulated circRNAs compared to linear RNA (**Extended Data. 8b&c**). Minimal *in vitro* innate immunogenicity and cell toxicity was observed for SEC purified circRNAs formulated by LPP.

We then evaluated the *in vivo* immunogenicity of RABV-G encoded RNA vaccines. By intramuscular injection into mice, it's proposed that the circRNA-LPP nanoparticles entered antigen-presenting cells *via* endocytic pathway and circRNA got released from endosome into cytoplasm for antigen translation. The translated antigens were then processed to elicit humoral and cellular immune responses as illustrated in **Fig. 5a**. Two doses of rabies vaccines were *i.m.* administered at an interval of two weeks in balb/c mice (n=5 per group). Mice were weighted after first dosing and no weight loss was observed for both circRNA and linear RNA group (**Extended Data. 8d**). Mice sera were collected on day 10, 14, 21 & 28 to assess the pseudovirus neutralizing antibody titer (pVNT) and spleens were extracted for cellular immunity assessment on day 28 (**Fig. 5a**). As neutralizing antibodies (nAbs) play a critical role in protection against rabies virus[39], monitoring the geometric titer (GMT) nAb level gave us a hint on the efficacy of rabies vaccines. In this study, linear mRNA encoding the same ORF (LR-Novo) as CR-Novo was used for comparison as well. On day 10 post-priming dose, LR-Novo elicited a slightly higher antibody titer compared to all circRNA groups possibly due to relatively faster onset expression of antigens, though overall titers remained low (**Fig. 5b**). Slower but durable antigen expression induced comparable immune responses among circRNA groups while CR1, with the lowest MFE, had 2-fold pVNT of LR-Novo and 4-fold pVNT of CR-Novo (**Fig.5c**). One week after booster (day 21), circDesign-optimized sequence groups (CR1-4) elicited much stronger humoral immune responses than the rest of groups (CR5-6, CR-Novo, CR-Ther and LR-Novo) (**Fig. 5d**). Of note, all circular RNA groups exhibited continued increase in pVNTs on day 28, whereas linear RNA had no further enhancement in antibody titer (**Fig. 5e**). The geometric increase (GMI) of pVNT was calculated based on the GMT on day 10 as baseline to show the trend of humoral immune response induction. As plotted in **Fig. 5f**, LR-Novo had the least GMI trend while CR1&3 showed durable induction of immune responses. The pVNT kinetics of each group as a function of time were summarized as **Fig. 5h**, suggesting the great potentials of circDesign-enabled vaccine design in triggering strong antibody responses. It's worth noting that cellular immune responses induced by circDesign optimized

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sequences (CR1-4) especially CR3 are more prominent than other groups (**Fig. 5g**). A higher frequency of RABV-G specific T cells secreting interferon γ (IFN-γ) was observed in CR3 group with a mean value of 280.1 spots counted using Enzyme-linked immunosorbent spot (ELISpot) assay (**Extended Data. 9**). Though the practical significance of higher RABV-G specific cellular immunity in rabies prevention remains elusive[40, 41], we speculate the improved T cell immune responses from circDesign optimized vaccine may play important roles in rabies virus clearance from the host[40]. Taking a closer look at the cellular immune activation profile, flow cytometry analysis of the RABV-G peptide stimulated T cells was performed. Following a stepwise gating of the cell population (**Extended Data. 10**), the ratio of $CD4^+$ and $CD8^+$ T cells producing IFN-γ, TNF-α, IL-2 and IL-4 were computed, respectively. The results indicated the strong RABV-G specific $CD4^+$ and $CD8^+$ T cell responses elicited by circDesign generated vaccines (**Extended Data. 11**). We observed higher ratios of IFN-γ, TNF-α and IL-2 secreting T cells but minimal IL-4 producing population in both $CD4^+$ and $CD8^+$ T cells (**Extended Data. 11**), suggesting a T_h1-biased immune responses induced by circRNA rabies vaccines.

Combined with the in-solution and in-cell stability results of naked or LPP formulated circRNA, these results convincingly demonstrated the improved potency from circDesign-optimized sequences owing to their well-defined structure and enhanced stability during storage and intracellular trafficking. The correlation analysis of *in vivo* pVNT and ELISpot on day 28 showed the relation to MFE of circRNA (**Extended Data. 12**).

The durable humoral immune response may be related to the *in vivo* stability of CR3

The abovementioned *in vitro* and *in vivo* evaluation of circRNAs implicated the importance of circRNA stability as well as translation efficiency in achieving optimal performance. Therefore, balancing the optimization of CAI (translation efficiency) and MFE (stability) with a well-defined circRNA folding structure might synergistically contribute to a top performer sequence, thus leading to maximized *in vivo* outcomes. Of all circDesign optimized sequences, CR3 showed superior stability, protein translation output, along with *in vivo* immunogenicity (See **Extended Data Table 1** for summarized assessment results). Compared to CR-Novo, it's intriguing to find that another benchmark sequence CR-Ther had lower protein translation and induced relatively weaker immune responses despite its higher CAI. Consequently, we further compared CR-Novo with CR3 regarding their *in vivo* stability to explain the advantages of circDesign-generated CR3 over CR-Novo in their potency and durability. To study the *in vivo* stability, LPP formulated CR3 and CR-Novo were *i.m.* administered into balb/c mice (n=5 per group). At 6, 24, 48, 72 and 168 h

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post-dosing, mice were sacrificed and total RNAs were extracted from different organs including muscle (injection site), lymph node, spleen, liver, lung, heart, kidney, brain and small intestine (sm. int.). A standard plasmid containing the target sequence was used as internal standard control to quantify the circRNA copies per μg of total RNA using qPCR analysis. The biodistribution results revealed the good safety profile of LPP delivery system, as major injected dose accumulated at the injection site without significant retention in liver both for CR3 (**Fig. 6b**) and CR-Novo (**Fig. 6a**). Moreover, as time went by, more fractions of circRNA were detected in spleen and lymph node where the immune activation mainly occurs. It's noted that CR3 had extended muscle retention proportion (88.5%, 70% and 29.4% at 24h, 48h and 72h, respectively) compared to CR-Novo (74.5%, 25.4% and 7.4% at 24h, 48h and 72h, respectively) (**Fig. 6c**). The comprehensive quantification data were summarized in **Extended Data Table 2**. It's hypothesized that higher *in vivo* stability of CR3 benefited the longer retention at injection site given the same dosage as CR-Novo. Within 72 h of administration, the circRNA quantification data of muscle at injection site suggested the higher abundance of CR3 (**Fig. 6d**). This potentially contributed to a sustained lymphatic migration of antigen presenting cells (APCs) to peripheral immune organs such as draining lymph nodes and spleen, leading to robust immune activation.

The long-lasting humoral immune response induced by circDesign optimized rabies vaccine

Above findings rendered CR3-LPP a more potent rabies vaccine candidate potentially inducing long-lasting adaptive immunity against rabies virus. To study the longevity of immune responses, balb/c mice (n=5 per group) were immunized with LPP formulated CR3 and CR-Novo with the same dosage (2 μg per immunization for prime and boost at an interval of two weeks). The mice sera were collected over a much longer period during which the nAb titer was monitored (**Fig. 6e**). As shown in **Fig. 6f**, CR3 induced a stronger humoral immune response following prime and booster. 14 days after booster dose (on day 35), the pVNT of CR3 group remained 1.7-fold of CR-Novo (**Fig. 6f**). With a relatively slower waning rate of pVNT, CR3 group exhibited 2.1-fold and 3.2-fold pVNT over CR-Novo group on day 90 and 163, respectively. According to the guidelines of rabies vaccine released by World Health Organization (WHO), a second booster was recommended when re-exposure to rabies virus occurred 3 months after the last booster[42]. Thus, we administered the second booster in mice on day 163 to evaluate the immune responses after re-boost. The second booster induced much stronger immune response in CR3 group with 4.6-fold and 3.9-fold pVNT of CR-Novo one week (day 170) and two weeks (day 177) after re-booster dose (**Fig. 6f**). Besides, the cellular immune responses were assessed by ELISpot on day 170 using mice spleenocytes. CR3 immunized mice had shown significantly stronger antigen-specific IFN-γ and TNF-α secretion, potentially leading to better

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clinic outcome.

Second antigen demonstrating the robustness of circDesign in optimizing circRNA for improved performance

The *in vitro* and *in vivo* results of circRNA encoding RABV-G demonstrated the usefulness of circDesign algorithm in circular RNA sequence design. Considering the potential applicability of circDesign for different circRNA design and optimization, we validated the robustness of this algorithm using varicella-zoster virus (VZV) glycoprotein E (gE) as a second antigen to develop circRNA-based vaccine. At the same time, a new self-cleaving ribozyme from *Azoarcus* pre-tRNA was adopted to mediate the circularization reaction (See Supplementary materials for sequence details).

Though with a different intron splicing, the circularization of circRNAs harboring circDesign optimized VZV gE sequences was still highly efficient. In this comparison, we included three control sequences: CV-Poka which stems from a CAI-maximized sequence (CAI = 1); CV-Ther from a public optimization tool supported by Thermo Fisher; CV-Ori refers to the original wild-type VZV strain sequence without optimization (**Extended Data Fig. 13a**). Non-denaturing agarose gel electrophoresis of circRNAs showed similar correlation of migration rate with structural compactness (**Extended Data Fig. 13b**). Circularization efficiency, thermodynamic stability and in-solution stability analysis of all circRNAs encoding VZV gE revealed a good correlation with MFE value and ensemble structure diversity (**Extended Data Fig. 13c,d&e**). As revealed by the secondary structure prediction and pairwise visualization (**Extended Data Fig. 14-17**), CV1-5 had well-defined structure and less IRES folding interruption. The protein translation level assessed by flow cytometry indicated the importance of stability factor in durable protein expression, highlighted by the MFI at 48 h post-transfection into HSkMC cells (**Extended Data Fig. 13f**). Furthermore, the *in vivo* immune responses triggered by LPP formulated circRNA VZV vaccines were evaluated in balb/c mice (n=5). CV1-5 elicited significantly higher binding antibody titers than benchmark circRNAs on day 21 and 28 (**Extended Data Fig. 18**). It was noteworthy that CV-Poka had compromised *in vitro* and *in vivo* performance despite of a maximized codon preference, in that its susceptibility to degradation as a result of high MFE value. Studies have suggested that eliciting a strong cellular mediated immune response is a key determinant for the vaccine protective efficacy against herpes zoster[43]. Therefore, we assessed the cellular immune response using ELIspot. It was inspiring to find that circDesign optimized sequences CV1-5 induced stronger VZV gE-specific T cells, which is a key determinant of VZV vaccine efficacy (**Extended Data Fig. 18d**).

Taken together, we successfully exploited the circDesign on the design of second

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antigen and validated the outstanding design strategy *in vitro* and *in vivo*.

Discussions

In this study, we devised a novel algorithm tool, called circDesign, for simplified circular RNA design to improve its stability as well as translatability. The usefulness of this circDesign platform was validated using RABV-G and VZV gE antigens against the benchmark designs which were designed via conventional CAI-oriented method. Here we present evidence supporting the enhanced stability and translation efficiency of circDesign optimized sequence which contributed to considerably stronger in vivo immune responses. We speculated that the sustained expression of antigens at high efficiency achieved greater antigen-specific immune activations, leading to distinguished vaccine-elicited humoral and cellular immune responses. Overall, these data demonstrated an approach for rational circRNA design in support of next generation vaccine and therapeutic development.

The major limitation of this study lies in the fact that we performed extensive search on specific ORF sequences without further optimization of other functional regions of circRNA such as IRES motif. To the best of our knowledge, the de novo design of an efficient IRES motif remains challenging as these virally or cellularly derived IRES motifs are usually sensitive to sequence manipulation including deletion, mutation, or insertion. The different domains of IRES sequence cohesively function to recruit PIC and assemble the intact ribosome complex only when these domains are well-folded. Therefore, we did not incorporate the modulation of IRES region in the circDesign at current stage. Provided that the structure-to-activity of IRES were elucidated at molecular level, it might be possible to realized bottom-up search or optimization of IRES sequence in circDesign algorithm.

Other than IRES-driven translation from circRNA, N6-methyladenosine (m6A) modification upstream of ORF has been reported to facilitate the translation as well as the stability[44, 45]. If chemically modified nucleotides were incorporated into circRNA, it would pose difficulties on our circDesign algorithm as the structure prediction based on unmodified nucleotides is not reliable. Given that the energy model of chemically modified nucleotide is available, we would be able to do extensive structure comparison for optimal sequence design.

Materials & Methods

Ensemble diversity calculation

Ensemble diversity (ED) measures the complexity of the structural ensemble, which can be computed from pair probabilities.

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$$\mathrm{ED} = \sum_{i,j} p_{ij} \times (1 - p_{ij})$$

Positional entropy calculation

Positional entropy (PE) is a metric that measures whether a specific nucleotide at position *k* predominantly occurs in the same configuration, paired or unpaired.

$$\mathrm{PE}(k) = -\sum_{i} p_{ik} \ln p_{ik}$$

Positional entropy of the IRES region is defined as: $\mathrm{PE(IRES)} = \sum_k \mathrm{PE}(k)$, where *k* is in the region of IRES. The positional entropy is 0 for a nucleotide that is always unpaired or always paired with the same partner. Thus, positions with low entropy are predicted with high confidence.

circRNA secondary structure prediction

The circRNA secondary structure was modeled by ViennaRNA[46] with parameters for circular (-p -d2 -c) RNAs. And the structure was visualized by RiboGraphViz with the cirRNA secondary structure from RNAfold.

Vector and plasmid construction

To construct a vector for circRNA synthesis, T7 promoter, homology arm, elements from permuted intron-exon (PIE) construct, spacer, IRES, and coding sequences were synthesized by General Biol and cloned into pUC57-mini plasmid. For linear RNA synthesis, the pUC57 plasmid vector containing T7 promoter, 5' UTR, Kozak sequence, coding region, 3' UTR and polyA tail was synthesized by General Biol. All the sequences coding Firefly Luciferase (FLuc), RABV-G and VZV-gE are supplied in supplemental materials.

Plasmid DNA synthesis and linearization

pDNA was transformed into *E. coli* (Tsingke Biotechnology) and cultured in 100 mL of Luria Broth (LB) with 100 mg/mL Ampicillin (Sangon Biotech, CN). Plasmid was purified using a NucleoBond Xtra Midi kit (MN) and the concentration and purity was measured on a NanoDrop One (Thermo Fisher). pDNA was linearized using SapI restriction enzyme for 3 h at 37 °C and purified as template. After sequencing verification, the products were stored at -20 °C and were used as templates for *in vitro* transcription (IVT).

In vitro transcription and circularization of RNA

For the chemically modified linear mRNA synthesis, *in vitro* transcription reaction with 100% N1-methylpseudouridine (Hongene Biotech) substitution was

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conducted at 30 °C for 16 hours. mRNA was co-transcriptionally capped using m7(3'OMeG)(5')ppp(5')(2'OMeA)pG capping reagent (Hongene Biotech) in a "one-pot" reaction followed by digestion with DNase I (Thermo Fisher). Linear mRNA was purified using Dynabeads MyOne Carboxylic Acid beads (Thermo Fisher).

For the circular RNA synthesis, The linear RNA precursor transcripts were produced using 1 μg of linearized DNA template in a MEGAScript™ reaction (Ambion) for 16 h at 30 °C, according to the manufacturer's protocol. Then RNA was treated with DNase I (Thermo Fisher) for 60 min at 30 °C, followed by lithium chloride precipitation. To obtain circularized RNA, GTP was added to a final concentration of 2 mM along with a buffer including magnesium (50 mM Tris-HCl, (pH 8.0), 10 mM $MgCl_2$, 1 mM DTT; Thermo Fisher). RNA was then heated to 55 °C and held for 45 min before purification using Dynabeads MyOne Carboxylic Acid beads (Thermo Fisher).

Liquid chromatography purification of circRNA

Circular RNA products were further purified *via* liquid chromatography to remove spliced introns and precursor linear mRNA using a 21.2×300 mm size-exclusion column (SEC) with a particle size of 5 μm and pore size of 1000 Å (Sepax Technologies). RNA was run on an AKTA avant 150 purifier system in RNase-free 150 mM phosphate buffer (pH = 6.0). Purified circRNA products were concentrated by lithium chloride precipitation and analyzed using Agilent 2100 bioanalyzer.

CircRNA sequence identification

To validate the circularization of RNA, the SEC purified RNA was reverse transcribed into cDNA using a HiScript III All-in-one RT SuperMix Perfect kit (Vazyme), followed by PCR with primers that can amplify transcripts across the splicing junction site. The PCR products were sequenced using Sanger sequencing to validate the back-splicing junction of the circular RNA.

Circularization efficiency comparison

To compare the circularization efficiency after circularizing reaction before SEC purification, two sets of primers were used in qPCR method. One set of primers was designed at 3 'intron, representing the part of RNAs that without successful circularization ; the other set of primers was designed at the junction site, representing the circRNAs formed by successful circularization. Therefore, the circularization efficiency of different RNAs was obtained by comparison, and the sequence of primers was as follows:

circ-Rabies-F (5'-3'): ACAAACGGCTATTATGCGTTACC;
circ-Rabies-R (5'-3'): GACTTGAACCCACACGACCG;
Linear-Rabies-F (5'-3'):AACGTCAAGACGAGGGTAAAGAGAG

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Linear-Rabies-R (5'-3'):GACTTGAACCCACACGACC
circ-VZV-F (5'-3'): TAGCTCATGCTTGGTTAGAGCAG;
circ-VZV-R (5'-3'): CCCCCTGGGAGTCGAACC;
Linear-VZV-F (5'-3'):CGGCACCCACCTAAGGCAAAC;
Linear-VZV-R (5'-3'):CGAACCCAGCACCAACGGATTC;

To calculate the Circularization efficiency of each sequence:

$$\frac{\text{power}[2, -(x-y)] \times 100\%}{\text{power}[2, -(x-y)]+1}$$

x: Ct of circular RNA, y: Ct of linear RNA

Cell culture & circRNA transfection

Human embryonic kidney 293 (HEK293, ATCC)/Human Skeletal Muscle Cells (HSkMC, ATCC) and Baby hamster kidney 21 (BHK-21, ATCC) were cultured in complete Dulbecco's Modified Eagle's Medium (DMEM, Gibco) containing 10% fetal bovine serum (FBS, Gibco) and 1% penicillin-streptomycin (Thermo Fisher). Cells were seeded in a 6-well plate at a density of 5×10^6 cells/well 24 h prior to transfection. Lipofectamine Messenger MAX (Thermo Fisher) was used according to the manufacturer's instructions for the transfection of circRNA.

Agarose gel electrophoresis of circRNA

To study the electrophoretic mobility profile of circRNA, RNA samples suspended in Ambion® RNA storage buffer (Thermo Fisher) were denatured at 75 °C for 5 minutes and snap –cooled on ice before loaded onto 1% agrose gel (130 V for 1 h at room temperature). DL5000 DNA marker (Vazyme) was used in the gel. Gel image was taken by Gel Doc XR+ Gel Documentation System (Bio-Rad).

In-solution stability assay of circRNA

To assess the In-solution stability of mRNA, samples were incubated in PBS buffer containing 10 mM Mg^{2+} . Sampling was conducted at different time points (0, 1, 2, 4, 8, 12, 16, 24, 32, 48 and 60 hours). RNA integrity was analyzed by Qsep100™ Capillary Electrophoresis System (BiOptic Inc). The integrity was represented as the proportion of full-length mRNA calculated on electropherogram. The data were normalized to time point 0 h.

To extrapolate the half-life of each sequence, one-phase decay equation:

$$Y = (Y_0 - \text{Plateau}) \cdot e^{-KX} + \text{Plateau}$$

was used to fit the curve. The Y_0 and Plateau were set as 100 and 0, respectively. Half-life was computed as ln(2)/K, where K refers to decay rate constant.

Thermodynamic stability assay of circular RNA

Equal volume of circRNA (1mg/mL) and diluted Quant-iT™ RiboGreen®

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(Thermo Fisher) reagent was mixed in 1× TE buffer. A total of 20 µL mixture was loaded into 384-well plate and centrifuged at 1000 rpm for 1 min before running the Melting Curve program on QuantStudio 5 Real-Time PCR System (Applied Biosystems). circRNA underwent a denaturing process at 95°C for 5 min with a ramp rate of 0.05°C/s starting from 25°C. Then the temperature stepped down to 25°C with a rate of 1.6°C/s for annealing. The fluorescent intensity data at different temperatures were normalized to the initial value and plotted in GraphPad Prism 10.0. Sigmoidal curve was used to fit the data to extrapolate the melting temperature (T_m) of each circRNA.

In-cell stability assay of circular RNA

HSkMC cells were seeded in a 6-well plate at a density of 5×10^6 cells per well 24 h prior to transfection. Lipofectamine Messenger MAX (Thermo Fisher Scientific) was used according to the manufacturer's instructions for the transfection of circRNA. Cells were collected and and RNA was harvested and purified at 3, 12, 24, 48 and 72 h after transfection using an RNeasy Mini Plus kit (QIAGEN). Synthesis of first-strand cDNA from total RNA was performed with HiScript® III All-in-one RT SuperMix Perfect for qPCR (Vazyme). Gene specific TaqMan primers (TsingKe) amplifying the circRNA junction site were synthesized as below:

circ-tzF (5'-3'): ACAAACGGCTATTATGCGTTACC;
circ-tzR (5'-3'): GACTTGAACCCACACGACCG;
circ-Probe (5'-3'): FAM-ACGGACTTAAAATCCGTTGAC-MGB

Target amplification sequence was inserted into plasmid as an internal standard to obtain the absolute circRNA copies from RNA extracts. The qPCR reaction was carried out using Probe qPCR Mix kit (Takara) . For each sample, threshold cycle values (Ct) were processed according to the comparative Ct method. Gene expression levels were presented relative to Standard curve.

In vitro innate immunogenicity assay

24 hours post transfection of circRNA into C2C12 cells (ATCC), cell culture supernatant was collected. Interferon Beta level in supernatant was assayed by VeriKine TM Mouse IFN Beta ELISA Kit (PBL assay science) according to manufacurer's instruction. Optical density (OD) was measured at 450 nm using Bio-Tek Synergy I plate reader. The concentration of IFN-β was deduced from the standard curve.

Flow cytometry

24 hours post transfection of circRNA, HSkMC cells were harvested and resuspended in 1 mL of PBS buffer at a concentration of 1×10^7 cells/mL. 100 µL of the resuspended cells was added to a FACS tube and stained with 100 µL of Live/Dead Fixable Aqua Dead Cell Stain (Thermo Fisher Scientific) at a 1:100 dilution and human FcR blocking reagent (Miltenyi Biotec) at a 1:20 dilution on

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ice for 30 min. Cells were then washed with 2.5 mL of FACS buffer and centrifuged at 1500 rpm for 5 min. After centrifugation, cells were stained with 1 µg (1:100 dilution) of a anti-Rabies Virus CVS-11 antibody (Merck) or anti-VZV gE mouse antibody (Abcam) at 4 °C for 30 min in dark before washing with 2.5 mL of FACS buffer and centrifuging at 1500 rpm for 5 min. Cells were then stained with 100 µL of Goat PE-anti-mouse IgG (1:100 dilution) (Abcam) at 4 °C for 30 min in dark. After incubation, cells were washed with 2.5 mL of FACS buffer, centrifuged at 1500 rpm for 5 min and resuspended with 250 µL of PBS. Samples were analyzed on BD Canto II (BD Biosciences). Data were processed using FlowJo V10.1 (Tree Star).

RNA sequencing and ribosome profiling of circular RNA

At 6 h post-transfection of CR-Novo and CR3 into HEK293 cells, cycloheximide (Sigma) was added to cell culture medium to a final concentration of 100 µg/ml to arrest protein synthesis and preserve ribosome-mRNA interaction. After a 2- min incubation at 4 °C and lysis, the cell lysate was centrifuged at 3,000×g and the supernatant containing free polysomes was decanted. To prepare ribosome footprints (RFs), RNase I (NEB) and DNase I (NEB) were added to lysate and was then incubated at room temperature on a Nutator mixer. Nuclease digestion was stopped by adding SUPERase·In RNase inhibitor (Ambion)followed by purification using size exclusion columns (GE Healthcare). Next, 10% (wt/vol) SDS was added to the elution, and RFs with a size greater than 17nt was isolated according to the RNA Clean and Concentrator-25 kit (Zymo Research). After obtaining ribosome footprints, Ribo-seq libraries were constructed using NEBNext® Multiple Small RNA Library Prep Set for Illumina® (catalog no.E7300S, E7300L). Briefly, adapters were added to both ends of RFs, followed by reverse transcription and PCR amplification. The 140-160bp size PCR products were enriched to generate a cDNA library and sequenced using Illumina HiSeqTM X10 by Gene Denovo Biotechnology Co. (Guangzhou, China).

Ribosome footprint data processing and alignment

Low quality reads were filtered by fastp. Raw reads containing over 50% of low-quality bases or over 10% of N bases were removed. Adapter sequences were trimmed. Reads with length between 20-40 bp were retained for subsequent analysis. Short reads alignment tool Bowtie2 was used for mapping reads to ribosome RNA (rRNA) database, GenBank, Rfam database. The reads mapped to rRNAs, transfer RNAs (tRNA), small nuclear RNAs (snRNA), small nucleolar RNAs (snoRNA), and miRNA were removed. Processed RNA-seq reads were aligned to the circular RNA using sequence alignment by STAR with 2-pass setting enabled. The reads were assigned to different functional regions (IRES, CDS, spacer, homology arm and exon) based on the position of the alignment. Translation efficiency (TE) was calculated as the base 2 logarithmic ratio of normalized RPFs over the normalized input for each ORF.

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Pause sites analysis

Because the translocation of ribosomes occurs along the mRNA one codon at a time, the potential of single-codon resolution pauses were investigated. Using PausePred (https://pausepred.ucc.ie/) which infers the locations of ribosome pauses, global single codon pauses were analyzed.

Quantification of read abundance

Reads counts in the different regions of circular RNA was calculated by software RSEM, and the level was normalized by using FPKM (fragment per kilobase of transcript per million mapped reads) method, and the formula is shown as follows:

$$FPKM = 10^6 C/(NL/10^3)$$

Given FPKM (A) to be the abundance of sequence A, C to be number of fragments mapped to sequence A, N to be total number of fragments that mapped to reference sequences, and L to be number of bases on sequence A. Therefore, the calculated FPKM can be directly used for comparing the difference of abundance among samples. The ribosome footprint depth was plotted in a circular form with each functional region denoted in different color.

Polysome profiling of circRNA

At 6 h post-transfection of CR-Novo and CR3 into HEK293 cells, cycloheximide (Sigma) was added to cell culture medium to a final concentration of 100 μg/ml to arrest protein synthesis and preserve ribosome-mRNA interaction, and the cells were incubated at 37 ° C for 15 minutes. After trypsin digestion (0.25%), harvest and wash cells with ice-cold PBS supplemented with 100 mg/mL cycloheximide. The frozen tubes were flash frozen in liquid nitrogen and then stored at -80℃ for later use. Samples were sent to NKY-GuangZhou Gene reader Bio-Medical Technologies Co., Ltd. (Guangzhou, China) for isolation. In brief, cells were lysed in lysis buffer (10 mM Tris-HCl pH 7.4, 150 mM NaCl, 5 mM $MgCl_2$, 1% Triton X-100, 0.5% deoxycholate, 2 mM DTT) supplemented with 4 U/mL RNase inhibitor, 100 μg/mL cycloheximide, and proteinase inhibitor cocktail (Roche, Basel, Switzerland) on ice for 20 min. After centrifugation at 13,000 g for 10 min, the supernatant was then taken and loaded into a 10% to 45% sucrose gradient solution and centrifuged at 360000 rpm for 3 h (Beckman Optima XE-100). 18 fractions from each sample were collected using a piston gradient fractionator (Biocomp, Fredericton, Canada) equipped with a Bio-Rad Econo UV monitor, followed by total RNA extraction from each fraction for RT-qPCR analysis.

Preparation of LPP-circRNA

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LPP nanoparticles were prepared by a two-step method as previously described[35]. Briefly, a cationic polymer compound (PbAE) was dissolved in PBS buffer solution (pH=5.2) and circRNA was diluted with citrate buffer（pH4, 5 mM ）and RNase-free water. mRNA/PbAE complexes were prepared by a microfluidic mixer (Inano D, Micro&Nano Technology Inc, China) at a volume ratio of 5:1 (mRNA: PbAE). After that, lipids were dissolved in ethanol at molar ratios of 40:15:43.5:1.5 (ionizable lipid: 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE):Cholesterol: PEG-lipid). The lipid mixture was combined with mRNA/PbAE complexes at a ratio of 3:1 (aqueous: ethanol) using a microfluidic mixer. Formulations were concentrated using Amicon ultra centrifugal filters (EMD Millipore) in sodium acetate with 10% sucrose buffer, passed through a 0.22-mm filter, and stored at 4℃.

Nanoparticle characterization

The size, polydispersity index (PDI) and zeta potential of LPP formulated circRNA were measured by dynamic light scattering (DLS) (Malvern, Zetasizer Nano ZS). Briefly, LPP-circRNA samples were diluted in sodium acetate with 10% sucrose buffer at a concentration of 0.1 mg/mL. The measurement was performed in triplicate with 8 runs in each measurement.

In vitro storage stability assay of LPP formulated circRNA

LPP formulated circRNA was stored in sodium acetate with 10% sucrose buffer at 4°C over a time course of 60 days. At Day 0, 30 & 60, LPP-circRNA was sampled for physicochemical characterization including mRNA content, hydrodynamic size and PDI.

Cell count Kit-8 (CCK-8) assay of LPP formulated circRNA

Cell proliferation capacity was evaluated using CCK-8 assay kit (Sigma Aldrich) according to the manufacturer's instructions. Briefly, HSkMC cells were seeded into 96-well plates at a density of 5×10^5 cells/well. On the other day, 100 ng of LPP formulated circRNA was added into cell culture directly. At 48 h post-transfection, 10 μL CCK-8 solution was added to each well and cells were incubated for a further 2 h at 37 °C. Optical density (OD) was measured at 450 nm using Bio-Tek Synergy I plate reader.

mRNA concentration and encapsulation efficiency of LPP-mRNA

LPP formulated circRNA was diluted with 1×TE buffer followed by mixing with an equal volume of 2% Triton X-100 and incubated at RT for 20 min to release the encapsulated mRNA-circRNA core. 100 μL of each sample was then transferred to the wells of a 96-well plate, to which 100 μL of 200 × diluted RiboGreen™ RNA Reagent (Thermo Scientific) was added. The plate was shaken and incubated at RT for 10 min. After that, the fluorescence of the plate was read by a Bio-Tek Synergy I plate reader (BioTek). A series of diluted standards were also

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prepared corresponding to 0.1–2.0 μg/mL mRNA. A standard curve of fluorescence as a function of mRNA concentration was plotted by linear regression, from which the mRNA concentrations of the samples were calculated. Encapsulation efficiency was calculated as Encapsulation efficiency (%) = [(total mRNA - free mRNA)/total mRNA] × 100%.

LPP formulated circRNA protection assay

Content analysis was performed to evaluate the integrity of the RNA after the formulation process and to assess RNase protection of the encapsulated RNA. For the RNase protection assay, 1 mAU of RNase A (Thermo Scientific) per microgram of RNA were added to the LPP/LNP/RNA formulation for 30 min at room temperature. RNase was then inactivated by incubating the sample with 1.68 mAU/μg RNA of proteinase K (Thermo Scientific) at 55 °C for 10 min. After RNase inactivation, residual RNA was detected from formulations (as described in the *mRNA concentration and encapsulation efficiency of LPP-mRNA* above).

Fluorescent microscopy imaging of circRNA expression

For the experiments where eGFP-mRNA was transfected, 5×10^5 HSkMC cells were incubated with 2 μg of eGFP-mRNA packaged in LPP or lipofectamine (Thermo Scientific) for 24 h in 6-well plate, and eGFP expression was visualized using an Eclipse DMi8 fluorescent microscope (Leica). Frequency of eGFP-expressing cells was determined using flow cytometry (BD FACSCanto II, Becton Dickinson). The excitation and emission was set at 488 nm and 525 nm, respectively.

In vivo biodistribution of FLuc encoded LPP-circRNAs in mice

Female balb/c mice aged 6-8 weeks were purchased from Shanghai Model Organisms Center. 5 μg of LPP formulated circRNA or linear RNA encoding Firefly Luciferase (FLuc) in PBS were administrated into mice (n=3 for each group) intramuscularly with 3/10 insulin syringes (BD biosciences). Mice were then anesthetized for whole-body bioluminescence imaging with an IVIS Spectrum (Roper Scientific) at different time points (6h, 24h, 48h, 72h, 96h and 192h).

Administration of LPP circRNA vaccine in mice

For immunogenicity studies, female balb/c mice (Shanghai Model Organisms Center) aged 6-8 weeks old (n=5 for each group) were used. 2 μg of LPP formulated circRNA vaccines were diluted in 50μL 1×PBS buffer and intramuscularly administrated into the mice's same hind leg for both prime and boost shots at an interval of two weeks. Mice in the control groups received PBS. The blood samples were collected on Day 10, 14, 21 & 28 post prime dosing.

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Mice spleens were also harvested at the endpoint (D28) for ELISpot and flow cytometry.

pVNT measurement

A pseudovirus neutralization assay was performed to evaluate the neutralizing ability of sera collected from immunized mice as described previously[ref]. In brief, serially diluted antibodies were first incubated with pseudotyped virus (Beijing Tiantan Biological) for 1 h, and the mixture was then incubated with BHK-21 cells. After a 24-h incubation in an incubator at 37 °C, cells were collected and lysed with luciferase substrate (PerkinElmer), then underwent luminescence intensity measurement by BioTek microplate reader. The IC50 value was calculated using 4 parameter logistic non-linear regression.

Enzyme-Linked ImmunoSorbent Assays (ELISA)

Briefly, recombinant VZV gE protein (Genscript) diluted in coating buffer (Biolegend) were used to coat 96-well EIA/RIA plates (Greiner Bio-one, 100 ng/well) at 4°C overnight. The plates were then washed with 1× PBS-T (0.05% Tween-20) and blocked with 2% BSA in PBS-T for 2 hours at RT. Mice sera samples with serial dilutions were added and incubated for 2 hours at RT. After washing, HRP-conjugated goat anti-mouse IgG Ab (1:10,000) was added and incubated for 1 hour. TMB substrate (Invitrogen) was then used for color development and the absorbance was read at 450 nm using BioTek microplate reader. Endpoint titers were calculated as the largest sample dilution factor yielding a signal that exceeds 2.1-fold value of the background.

In vivo mouse cytokine analysis

A serial dosage of LPP formulated CR3, CR-Novo and LR-Novo (0.02 μg, 0.2 μg, 2 μg) were intramuscularly administered into female Balb/c mice (n=5, Shanghai Model Organisms Center) aged 6-8 weeks old. Mice sera were collected retro-orbitally at 4 h and 7 days post administration and subjected to cytokine analysis (Cytometric Bead Array Mouse Inflammation Kit, BD) using flow cytometry following the manufacturer's instructions. Data were presented as the percentage change relative to baseline mice group administered with PBS.

IFN-γ ELISpots

Assessment of the IFN-γ T cell response was performed using the Mouse IFN-γ ELISpotPLUS kit (Mabtech) following the manufacturer's instructions. Briefly, anti-IFN-γ pre-coated plates were blocked with DMEM+10% FBS for at least 30 min, then cells were added at 3×10^5 cells per well for negative control (media only) and RABV-G or VZV-gE peptide pools (15-mers overlapping by 11; Genscript) (1 μg/mL) in 200 μL final volume per well. The positive control wells contained 5×10^4 cells per well in 200 μL final volume per well with 5 μg/mL of

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ConA. Plates were incubated overnight at 5% CO_2, 37 °C incubator and developed as per the manufacturer's protocol. After overnight stimulation, plates were washed and sequentially incubated with biotinylated IFN-γ detection antibody (R4-6A2), streptavidin-ALP, and finally BCIP/NBT. Plates were imaged with ImmunoSpot Analyzer and quantified with ImmunoSpot software.

Biodistribution of LPP-circRNA

The mice were divided into two groups, and 5 μg CR-Novo-LPP or CR3-LPP was injected into the hind legs, respectively. 5 mices in each group were euthanized at 2h, 6h, 24h, 48h, 72h and 168h after administration. Tissue samples including heart, liver, spleen, lung, kidney, small intestine, brain, muscle at the injection site, and lymph node were collected and rapidly frozen in liquid nitrogen, and RNA was extracted using trizol method. The levels of CR-Novo or CR3 in tissue samples were analyzed by RT-qPCR. GraphPad Prism 8 and Origin2021 software were used for data processing.

Gene specific TaqMan primers amplifying the circRNA junction site were synthesized (TsingKe) as below:

circ-tzF (5'-3'): ACAAACGGCTATTATGCGTTACC;

circ-tzR (5'-3'): GACTTGAACCCACACGACCG;

circ-Probe (5'-3'): FAM-ACGGACTTAAAATCCGTTGAC-MGB

Target amplification sequence was inserted into plasmid as an internal standard to obtain the absolute circRNA copies from RNA extracts. The qPCR reaction was carried out using Probe qPCR Mix kit (Takara). For each sample, threshold cycle values (Ct) were processed according to the comparative Ct method. Gene expression levels were presented relative to Standard curve.

Correlation analysis

The Pearson correlation coefficient was calculated to assess the relationship between continuous variables, for example, experiment half-life and MFE.

Statistical Analyses

Two-tailed Student t test was used to determine statistical significance for data comparisons at a single time point. One-way ANOVA was used to determine statistical significance for data comparisons with multiple time points. Prism version 8.0 (GraphPad) or Origin 10.0 were used for generation of all graphs and performance of statistical analyses. Statistical significance is denoted as ns, not significant, $*p < 0.05$, $**p < 0.01$, and $***p < 0.001$.

Data Availability

The RNA sequences and DNA primer sequences used in the biological experiments are included in the Supplementary Sequence Information section (at the end of the manuscript). and are also uploaded in the submission system. We will release data at a permanent location upon acceptance and the accession codes

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will be available before publication.

Code Availability

The code has been submitted as part of this submission and will be released at a permanent location upon acceptance.

Conflict of Interest

L.H. is the founder of Coderna.ai. C.X., W.W., Y.T., H.S., Y.Y., Q.W., H.S., and H.L. are employees of StemiRNA Therapeutics. C.X., W.W., Y.T., H.S., and H.L. have filed provisional patents for the circRNA rabies and VZV vaccines.

Author Contributions

H.L. conceived and directed the project. H.L., H.S. and C.X. supervised all the *in vitro* and *in vivo* experiments. L.Z. and L.H. developed the circDesign algorithm to design circRNA sequences. C.X., W.W., X.W. and J.A. performed the mRNA synthesis and gel electrophoresis experiments. H.X., X.Q. and M.S. performed the liquid chromatography analysis and purification. C.X., W.W., J.A., X.S., Y.Y. and H.S. performed the protein expression and *in vivo* assays, and Q.W. conducted LPP formulation and chemical stability assays. Y.T. performed the ribo-seq experiments. H.Z. and Y.G. performed the animal study including injection and *in vivo* imaging. C.X., L.Z., H.L., H.S., W.W., Y.T., Q.W., and L.H. wrote the manuscript.

Acknowledgements

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Figure Legends

Figure 1. circDesign approach overview and design of circular RNA molecules encoding RABV-G. a, circDesign workflow illustration from library screening to structure-based analysis. Given a specific amino acid sequence, mRNA sequence pool was generated followed by candidate screening based on MFE and CAI. By incorporation of ORF into circRNA backbone, the circRNA folding, ensemble

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diversity and cross-region interaction was assessed. The designed circRNAs were then evaluated in vitro and in vivo. b, Distribution of circDesign-generated ORFs (R1-R6) and benchmark sequences (Ther & Novo) in a two-dimensional plot. The color darkness indicates the ensemble diversity. Schematic illustration of circRNA sequence designs composed of different functional regions for circularization. Homo arm refers to homology arm for 5'- & 3'- end pairing. c, Representative circularized diagram showing the CR3 and CR-Novo sequence. d, Secondary structure of CR3 and CR-Novo (uncolored: IRES motif; red: loop structure; blue: stem structure). e, Pairwise visualization of CR3 and CR-Novo showing the intra- and inter-region interaction (red: inter-region interaction; blue: inter-region interaction). Each region was colored as indicated.

Figure 2. Characterization of circular RNA molecules encoding RABV-G. a, Comparison of the circularization efficiency of different circRNA using qPCR method. Data in triplicate were presented as mean ± s.d. b, Native agarose gel characterization of the circRNA showing the correlation of gel mobility with circRNA folding compactness. c, Thermodynamic stability of circRNA molecules suggesting a correlation of MFE with melting temperature. The mean values of three independent measurement data were fitted with sigmoidal curve to extrapolate the melting temperature. d, In-solution degradation curve of circRNA upon incubation in PBS buffer containing 10 mM Mg^{2+} at 37 °C. Triplicate data were presented as mean ± s.d. and fitted with one-phase decay curve to obtain half-life and decay constant. e, In-cell degradation of all circRNAs within 72 h post-transfection in HSkMC cells. Triplicate data points were presented as mean ± s.d. followed by non-linear one-phase decay fitting. The original value at the 0 h was set as 125% with a final plateau value at 0%. f, Flow cytometry analysis of circRNA in comparison with linear RNA in HSkMC cells at 24 h post-transfection. Mean fluorescent intensity (MFI) of protein expression level from cells transfected circRNA or linear mRNA was calculated. Three independent measurement data were plotted as mean ± s.d. One-way ANOVA with multiple comparisons to LR-Novo group was performed for statistical analysis. g, Correlation analysis of MFE with experiment in-solution and in-cell half-life. h, Correlation analysis of predicted half-life with experiment half-life.

Figure 3. Translation efficiency comparison of CR3 and CR-Novo. Predicted secondary structure of a) CR3 and b) CR-Novo. c, Simplified workflow diagram of ribosome profiling and polysome profiling of circRNA. d, Abundance of Ribo-seq and RNA-seq counts of circRNA (CR3 & CR-Novo) in transfected cells showing the higher translation efficiency of CR3. d, Total RNA-seq and Ribo-seq

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fpkm of CR-Novo and CR3 showing the relatively higher RNA abundance and higher ribosome footprint counts in cellular context. e, circular barplot showing the ribosome footprints of CR3 and CR-Novo. h, Polysome profiling results of CR3 and CR-Novo. f, Total polysome profiling of CR3 (black) and CR-Novo (red) transfected cells. g, RT-qPCR quantification of CR3 (black) and CR-Novo in collected ultracentrifuge fractions. The colored region refers to different major fractions from 40S, 60S, 80S (monosome) to polysomes. Data in triplicate measurement were presented as mean ± s.e.m.

Figure 4. Characterization of LPP formulated circRNAs. a, Scheme of two-step manufacturing of LPP formulated circRNA. The lipid mixture contains ionizable lipid, PEGylated lipid, cholesterol and helper lipid. b, The hydrodynamic size and polydispersity index (PDI) of LPP formulated linear RNA (LR-Fluc) and circular RNA (CR-FLuc). Zeta potential of LPP formulated circRNA and linear RNA. Three independent measurement data were plotted as mean ± s.e.m. d, Representative whole-body IVIS images of mice administered LPP CR-FLuc or LR-FLuc at different time points (6h, 24h, 48h, 72h, 96h and 168h). e. Total flux curve of luminescence as a function of time in balb/c mice (n=3; Green: CR-FLuc-LPP; Pink: LR-FLuc-LPP; Grey: PBS). f, Area under the curve (AUC) calculation of three groups from 6 h to 168 h suggesting the accumulated luciferase activity over the time course. Each dot represented the mean ± s.e.m. of data from three mice. Statistical significance was determined by One-way ANOVA multiple comparisons with LR-FLuc, ***$p < 0.001$; **$p < 0.01$; *$p < 0.05$; ns, not significant. g, Hydrodynamic size and PDI of LPP formulated circRNA and linear RNA. Three independent measurement data were plotted as mean ± s.e.m. h, Zeta potential of LPP formulated circRNA and linear RNA. Three independent measurement data were plotted as mean ± s.e.m. i, Storage stability of LPP formulated circRNA and linear RNA at 4°C implying the shelf life of circRNA vaccines. Samples in triplicate were treated and tested. The data to each sample were normalized to its D0 as 1 and plotted as mean ± s.e.m.

Figure 5. In vivo immune responses induced by circRNA rabies vaccine. a, Balb/c mice (n=5) immunization schedule and proposed mechanism of action of LPP circRNA rabies vaccines. Pseudovirus neutralizing titer (pVNT) of mice sera immunized with two doses of LPP formulated RNA vaccines on b) D10, c) D14, d) D21 and e) D28. pVNT data were presented as geometric mean ± geometric s.d. f, The geometric increase (GMI) of pVNT from mice vaccinated with different RNA vaccines using D10 pVNT as the baseline. GMI data were plotted with geometric s.d. g, IFN-γ ELISpot analysis of mice splenocytes on Day 28 showing

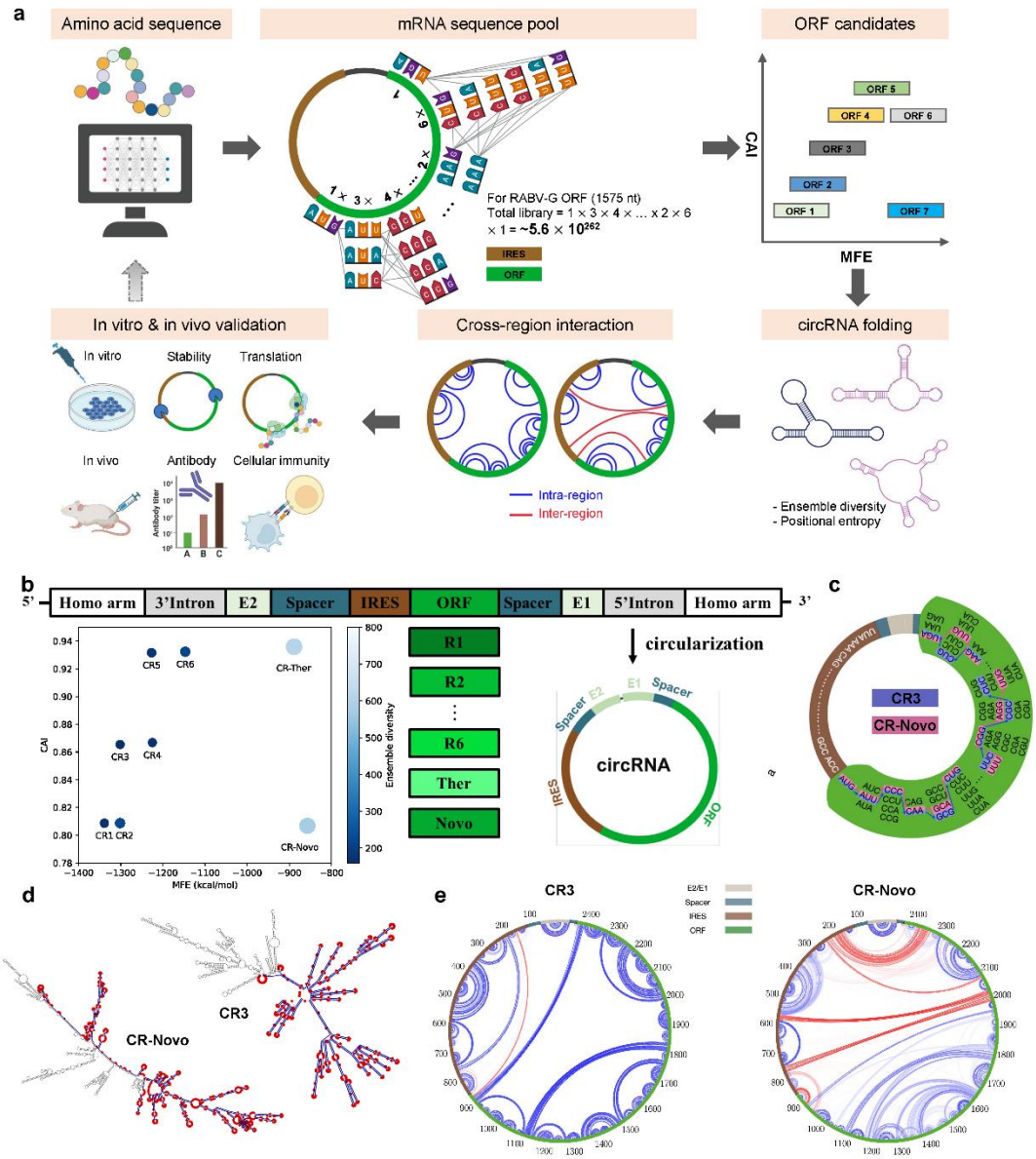
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robust cellular immune responses triggered by circDesign optimized sequences. Data in graph represented as s.f.c. per 3*10^5 cells. Each point on violin plot represents a single mouse. Bold black line represents the median. The significance between groups was determined using a one-way ANOVA with Dunn's multiple comparison test, ***p < 0.001; **p < 0.01; *p < 0.05; ns, not significant. s.f.c., spot-forming cells. h, Summary of pVNT results from D10 to D28 of all vaccinated balb/c mice (n=5) Data were presented as geometric mean ± geometric s.d.

Figure 6. In vivo biodistribution and immune response longevity analysis of LPP formulated rabies vaccines. In vivo biodistribution of LPP formulated circRNA. RT-qPCR quantification of a) CR-Novo and b) CR3 in different tissues at different time points (6h, 24h, 48h, 72h and 168h) in balb/c mice (n=5 per group at each time point). c, Pie plot showing the tissue distribution of circular RNA at various time points. d, Parallel comparison of circRNA abundance in muscle tissue as a function of time. Statistical significance was determined by multiple Mann-Whitney u tests ***p < 0.001; **p < 0.01; *p < 0.05; ns, not significant. e, Scheme of immunization schedule in balb/c mice study (n=5). f, Pseudovirus neutralizing antibody titer kinetics of CR3 and CR-Novo. Data were presented as geometric mean ± geometric s.d. Statistical significance was determined by multiple Mann-Whitney u tests ***p < 0.001; **p < 0.01; *p < 0.05; ns, not significant. g, IFN-γ and TNF-α ELISpot analysis of mice splenocytes on Day 28 showing robust cellular immune responses triggered by CR3. Data in graph represented as s.f.c. per 3*10^5 cells. Each point on violin plot represents a single mouse. Bold black line represents the median. The significance between groups was determined using a one-way ANOVA with Dunn's multiple comparison test, ***p < 0.001; **p < 0.01; *p < 0.05; ns, not significant. s.f.c., spot-forming cells.

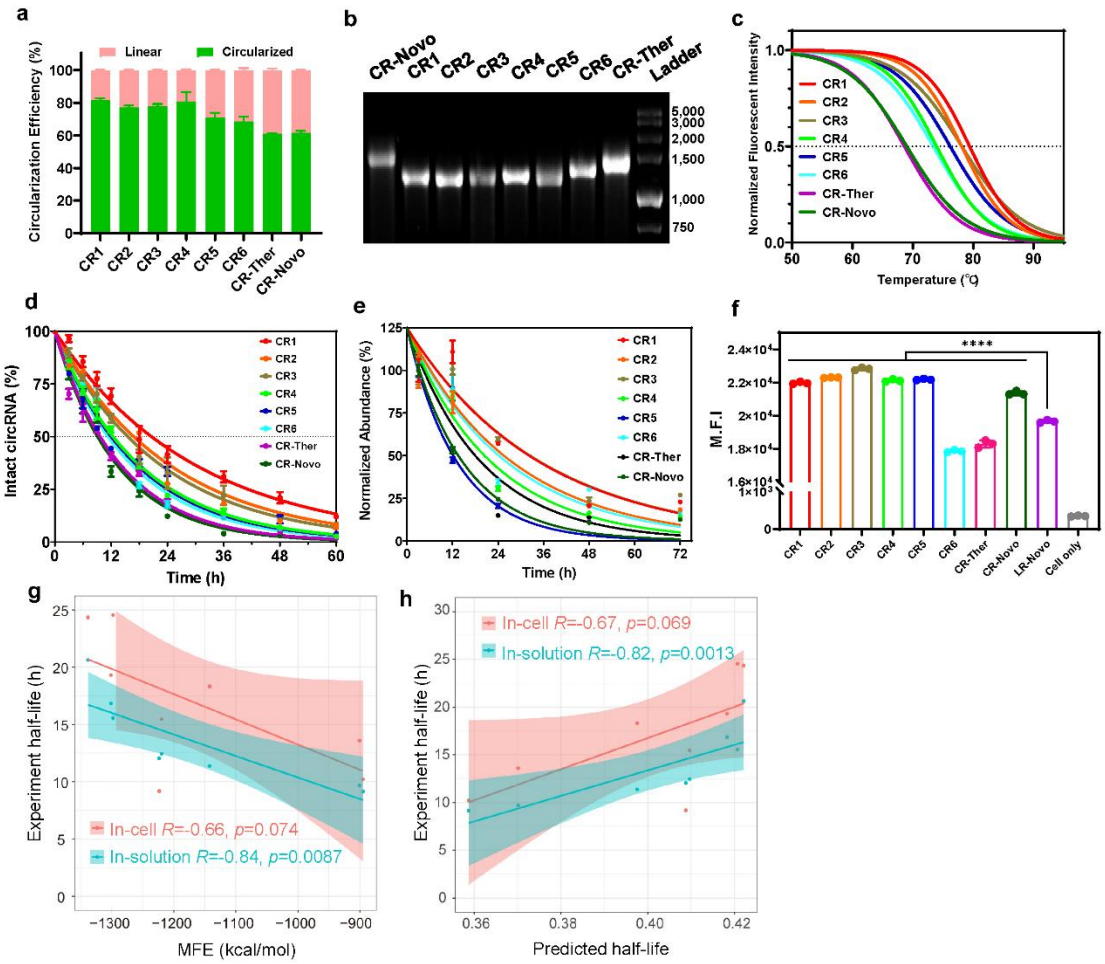
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Figure 1



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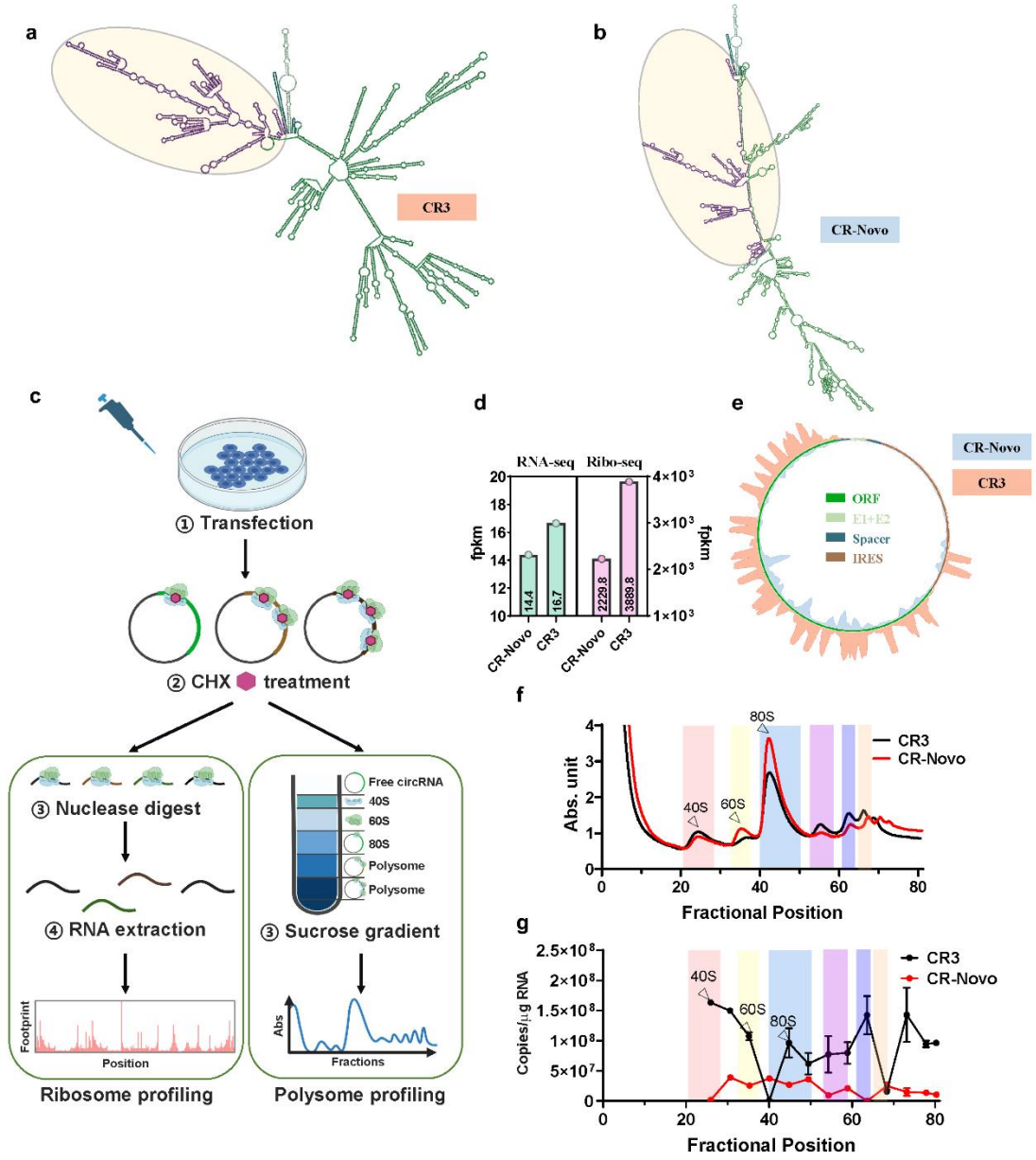
Figure 2



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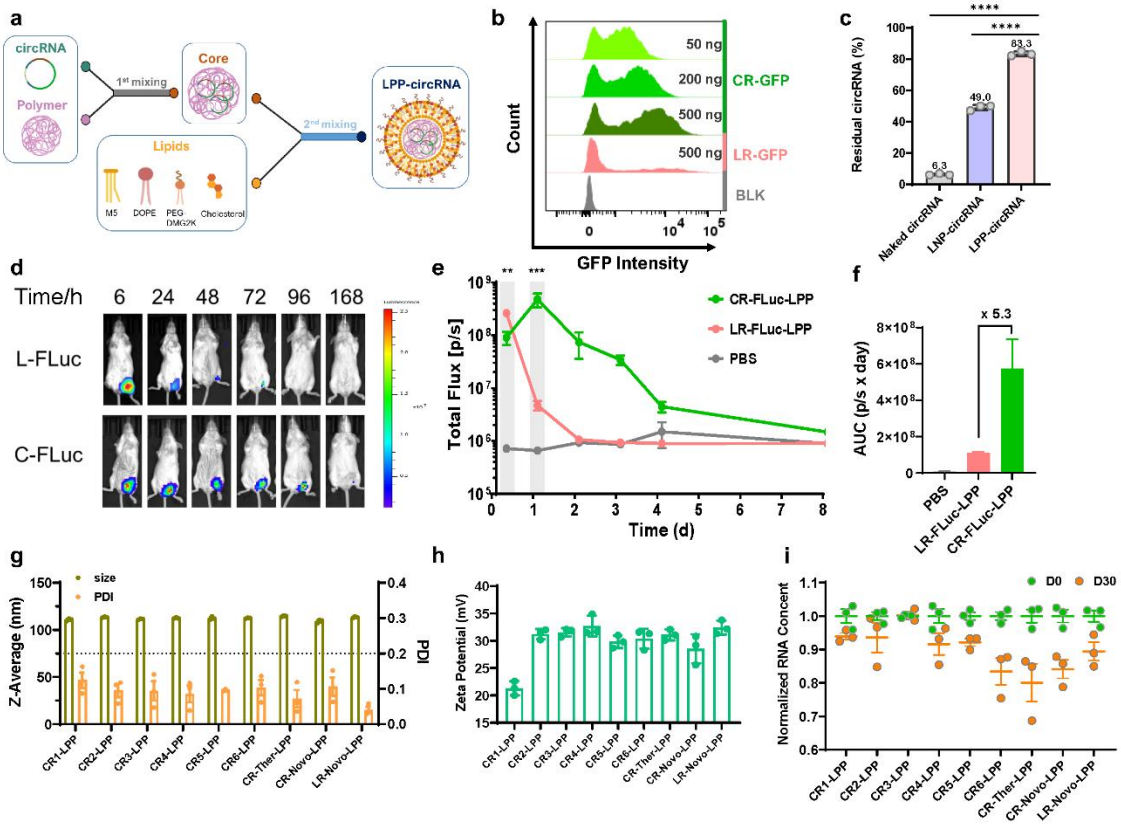
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Figure 3



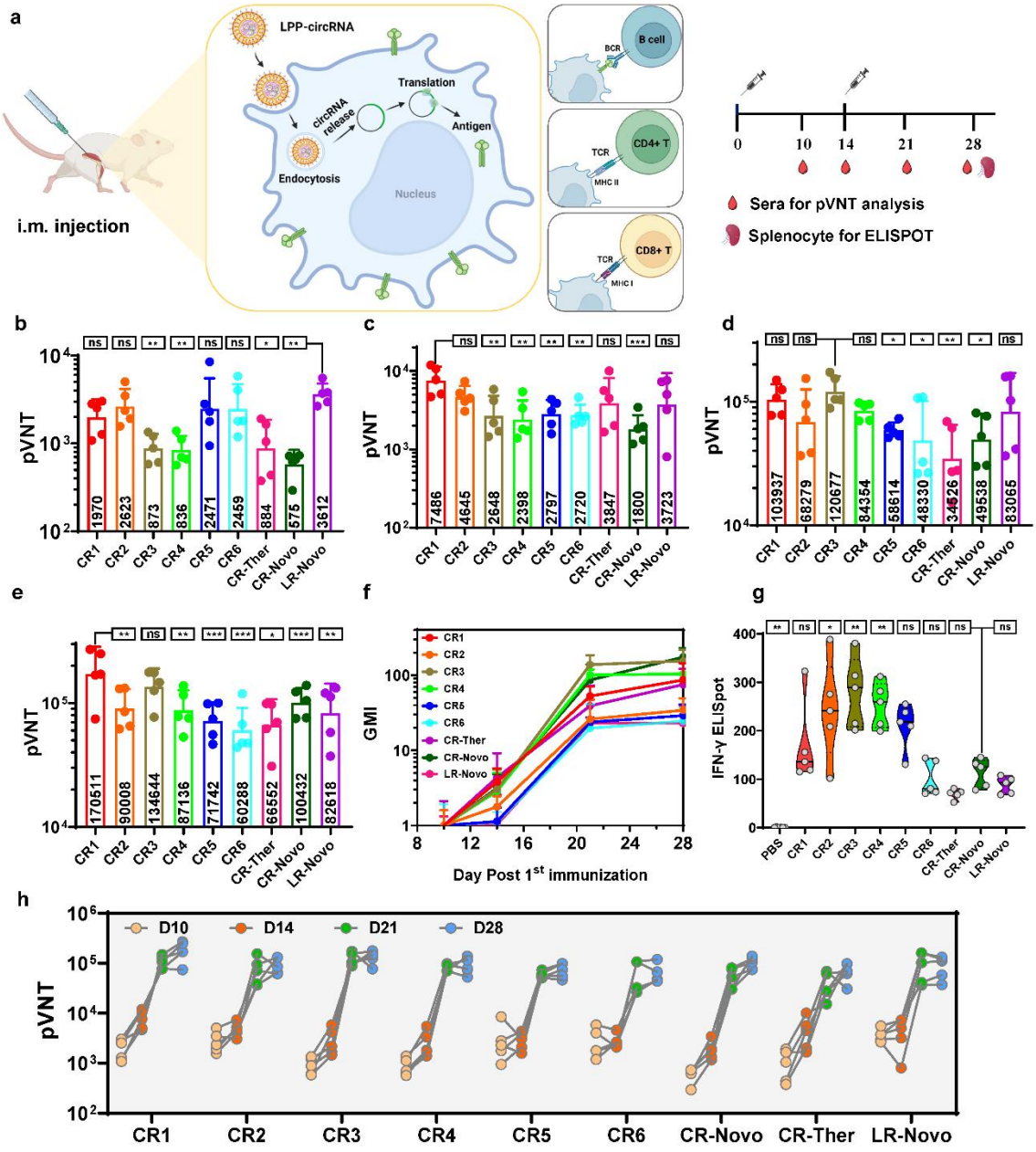
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Figure 4



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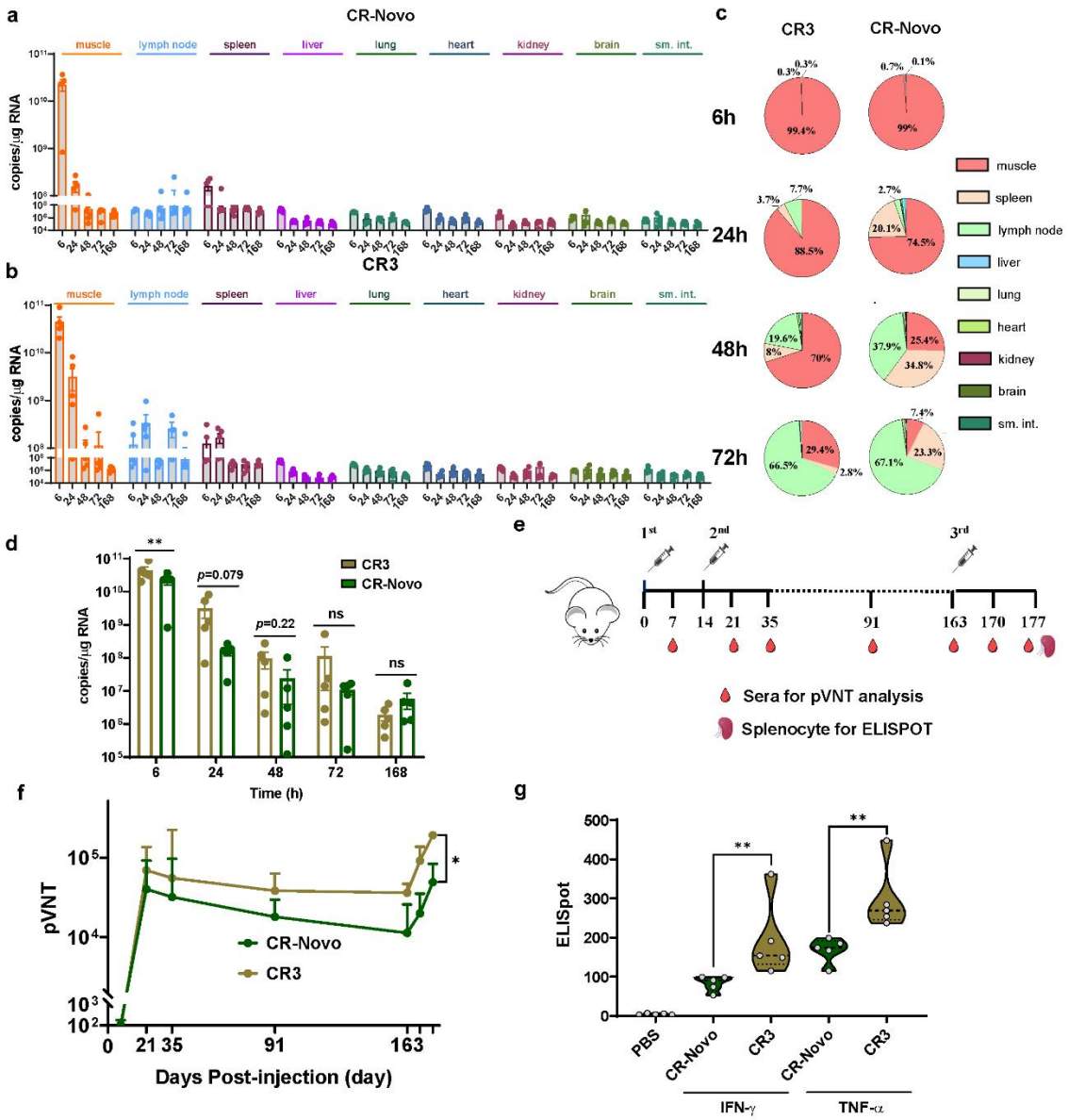
Figure 5



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Figure 6



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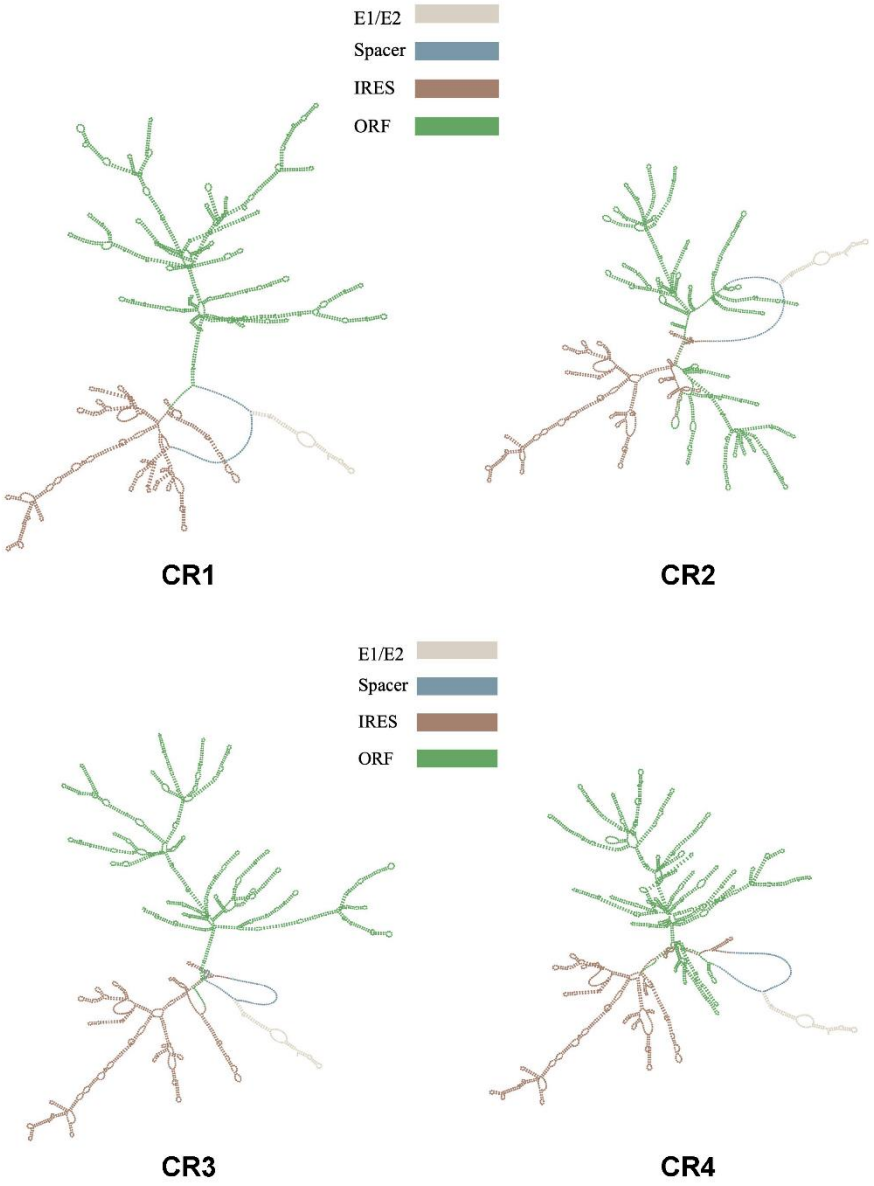
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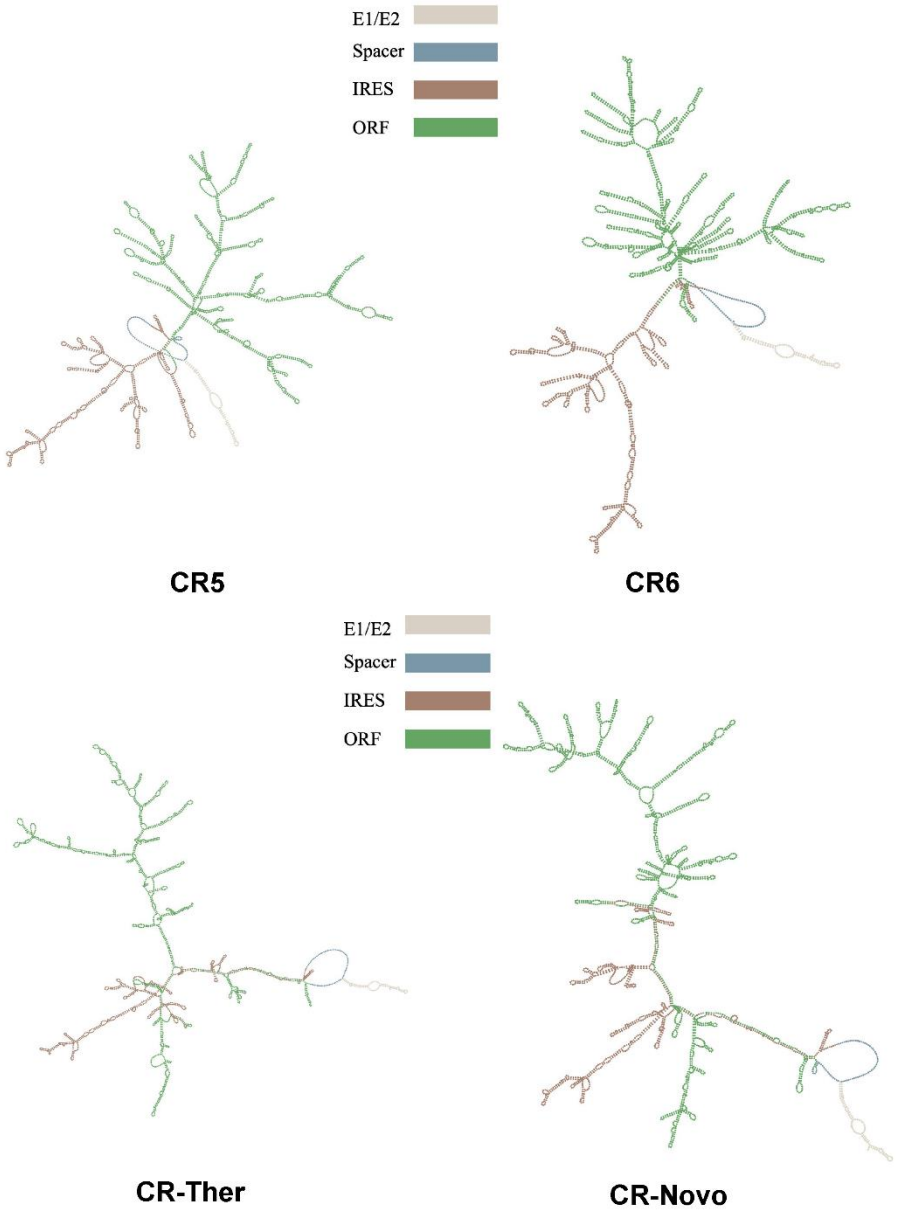
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Extended Data Figure 1



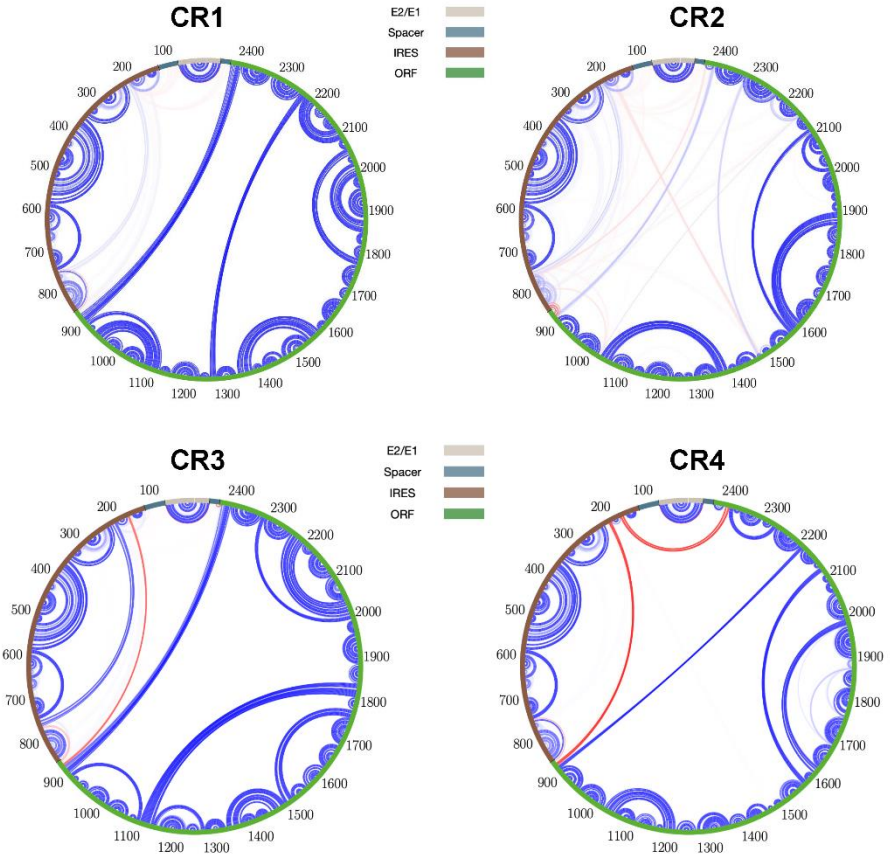
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Extended Data Figure 2



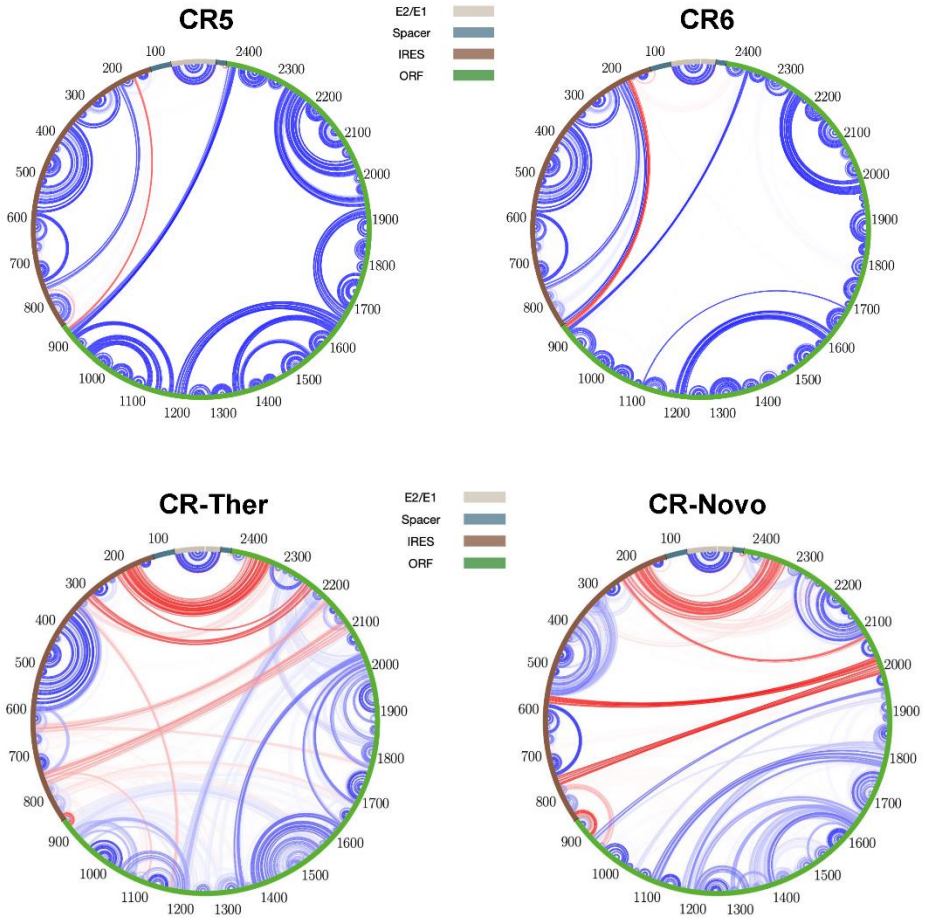
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Extended Data Figure 3



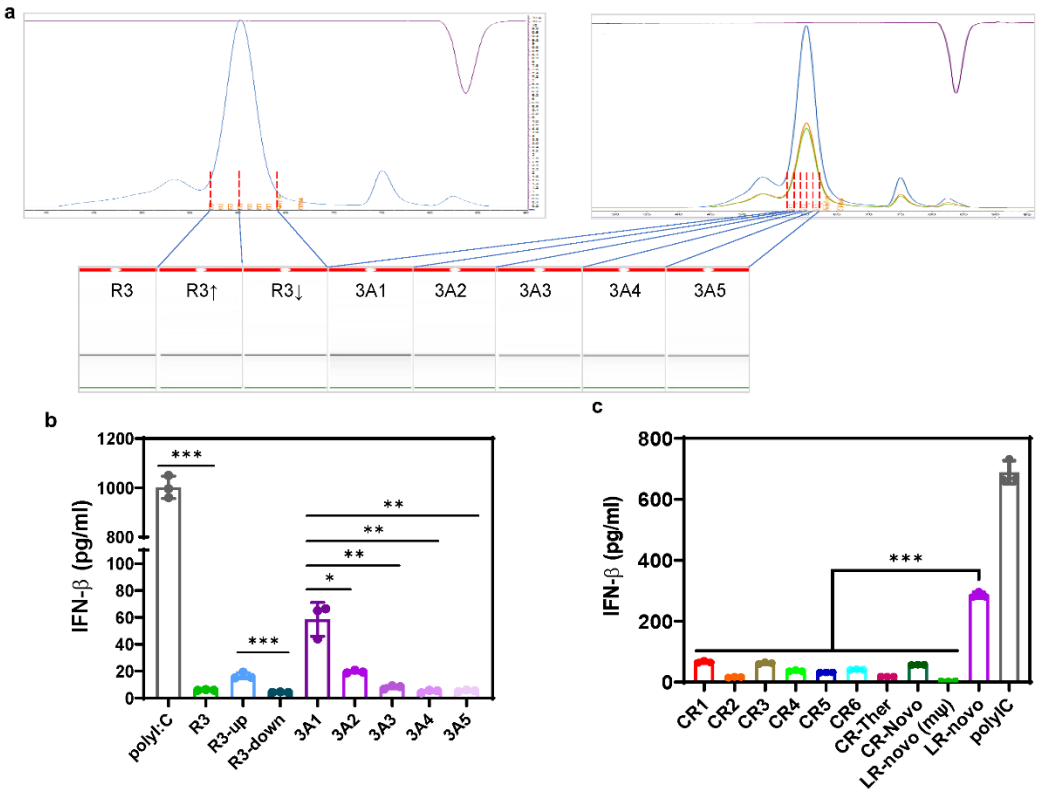
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Extended Data Figure 4



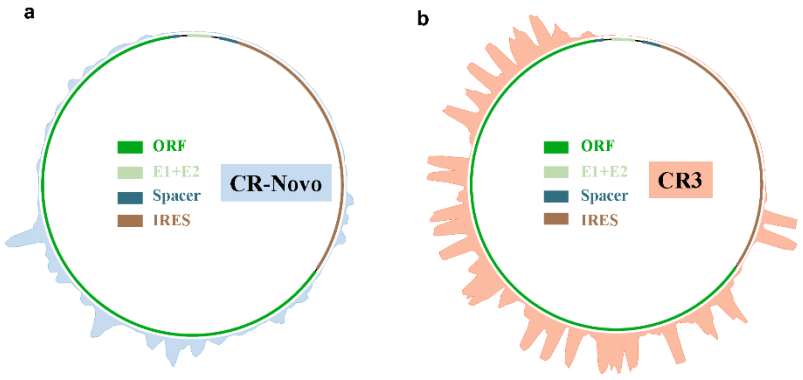
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Extended Data Figure 5



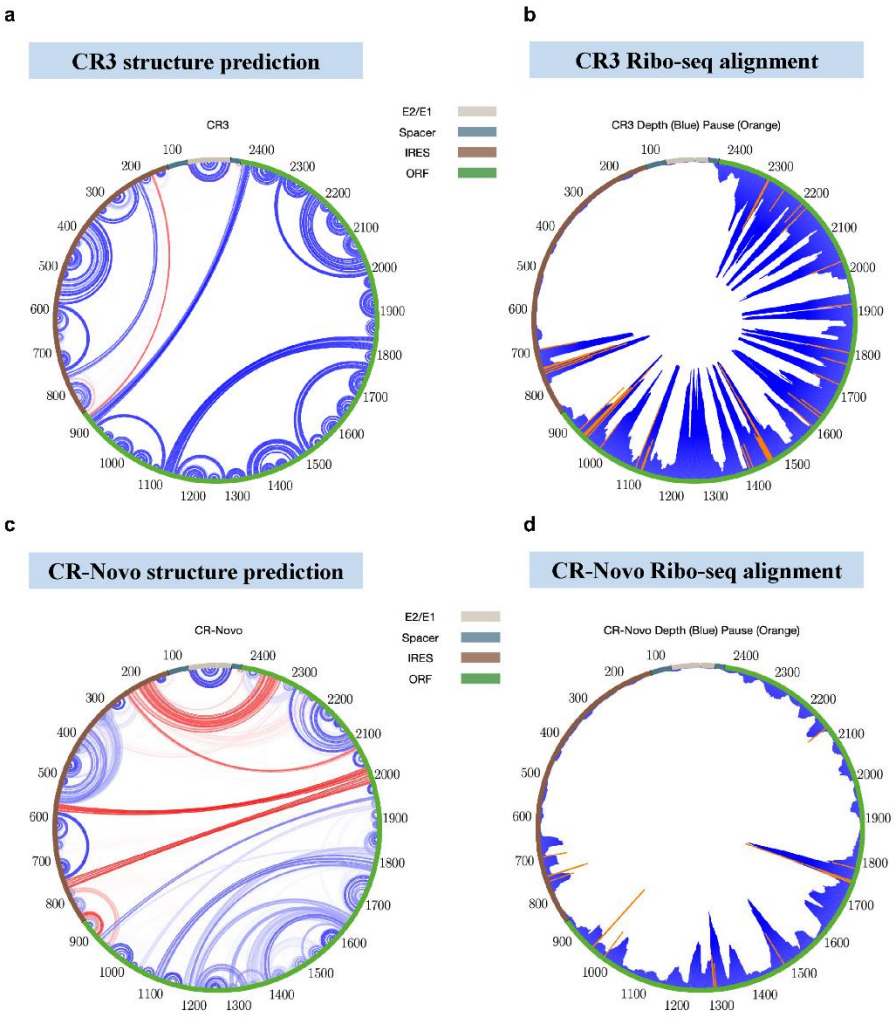
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Extended Data Figure 6



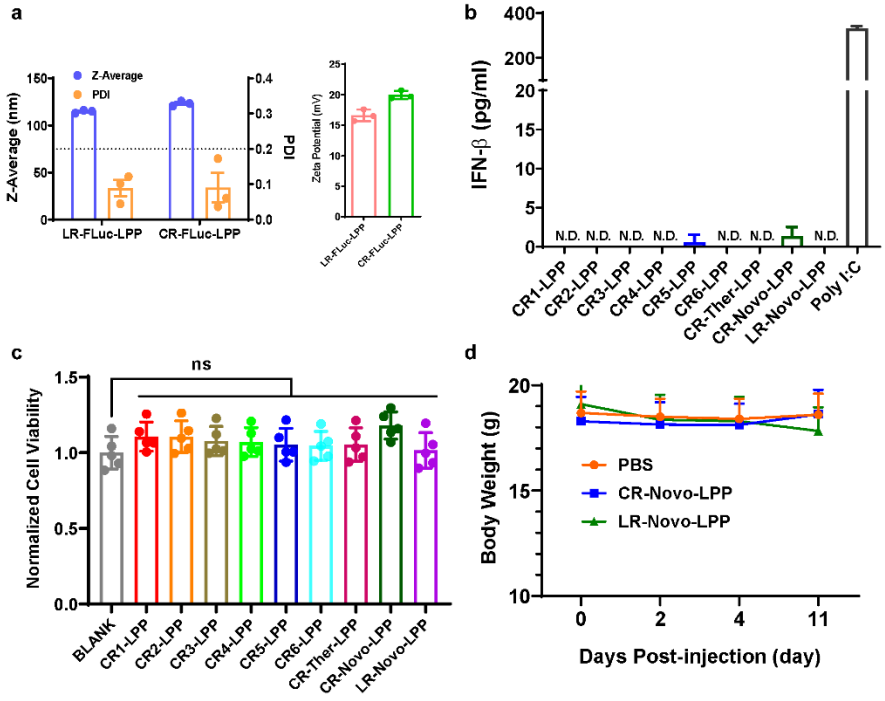
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Extended Data Figure 7



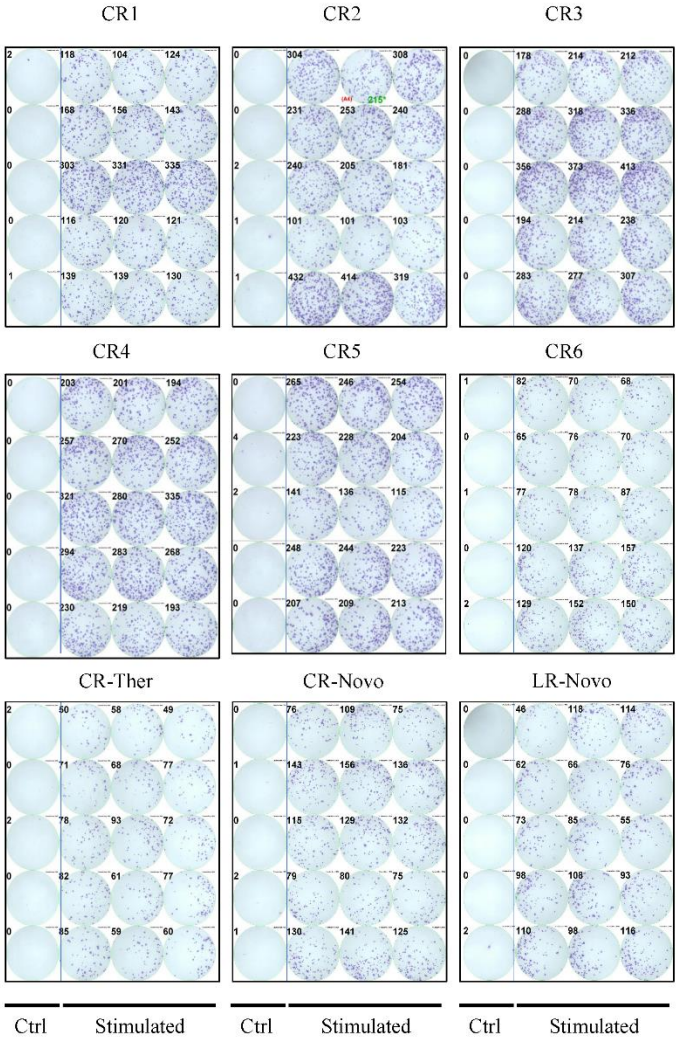
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Extended Data Figure 8



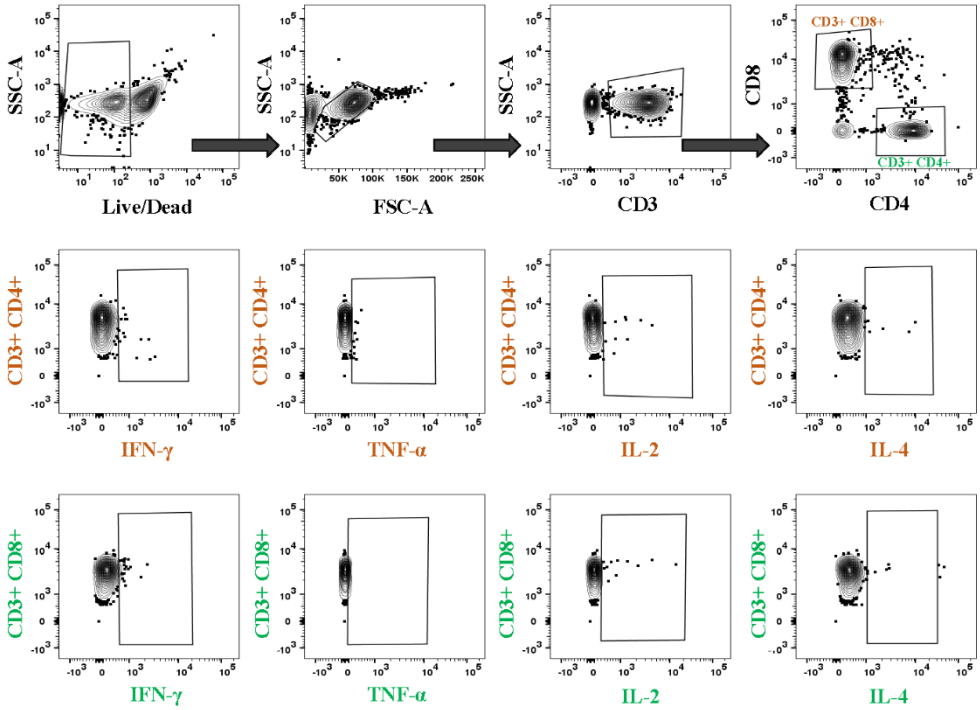
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Extended Data Figure 9



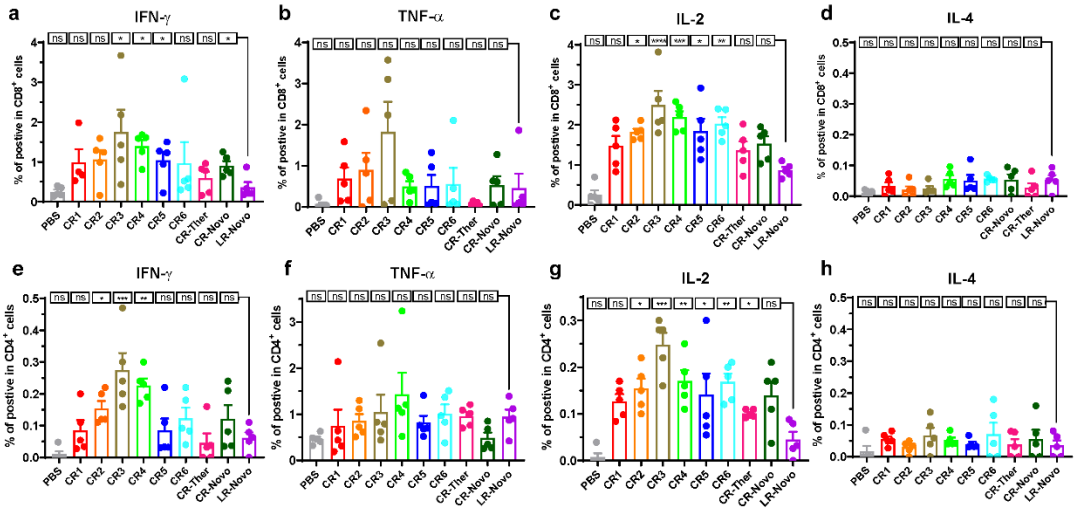
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Extended Data Figure 10



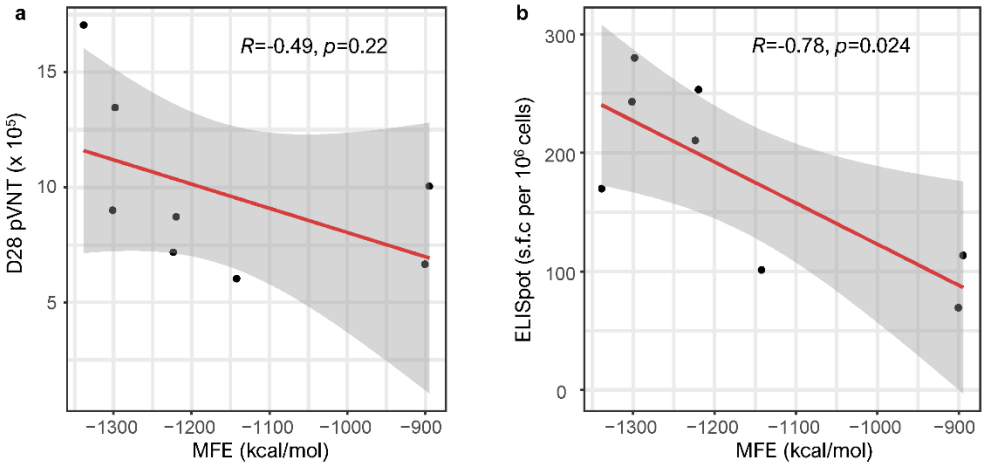
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Extended Data Figure 11



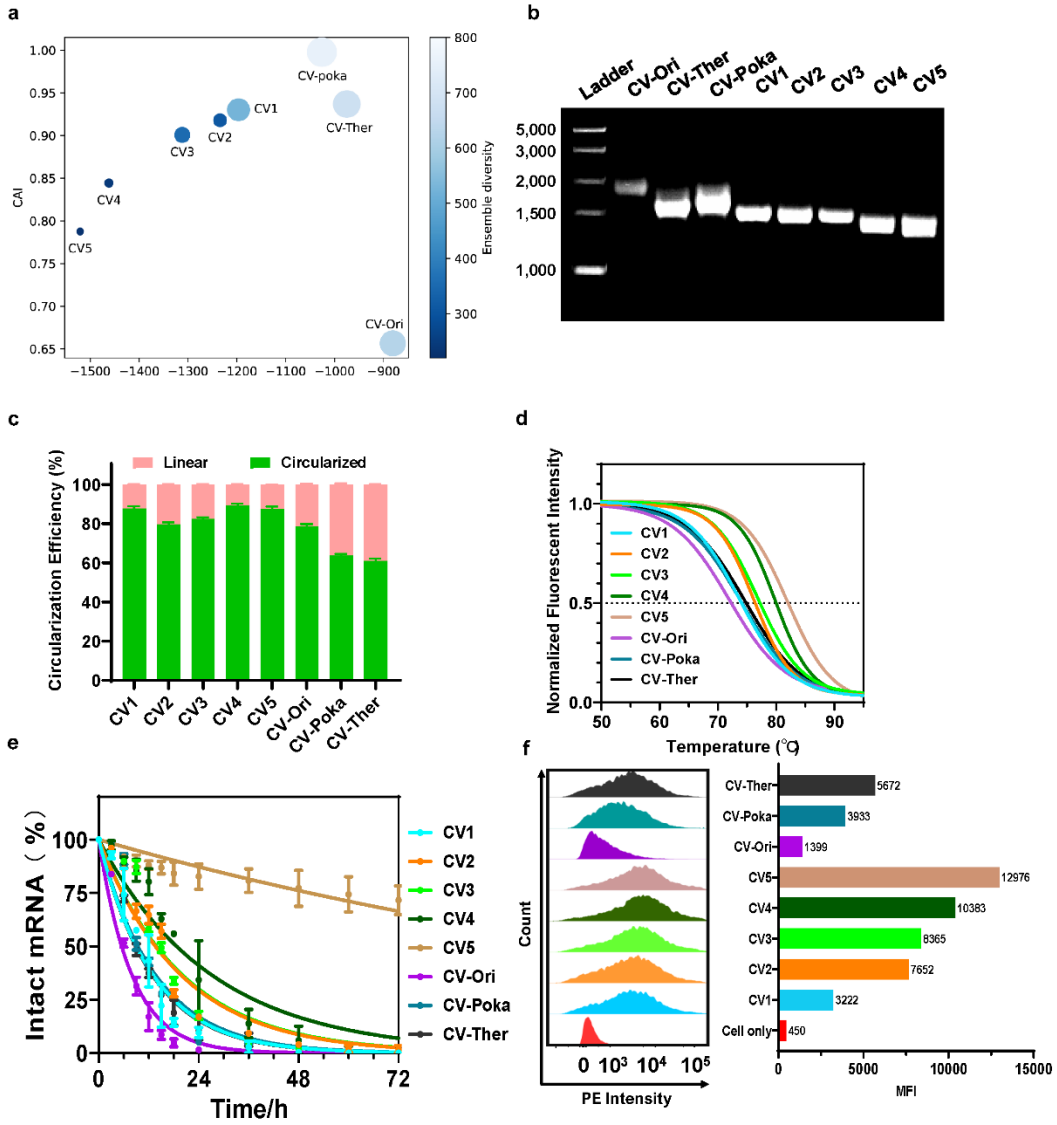
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Extended Data Figure 12



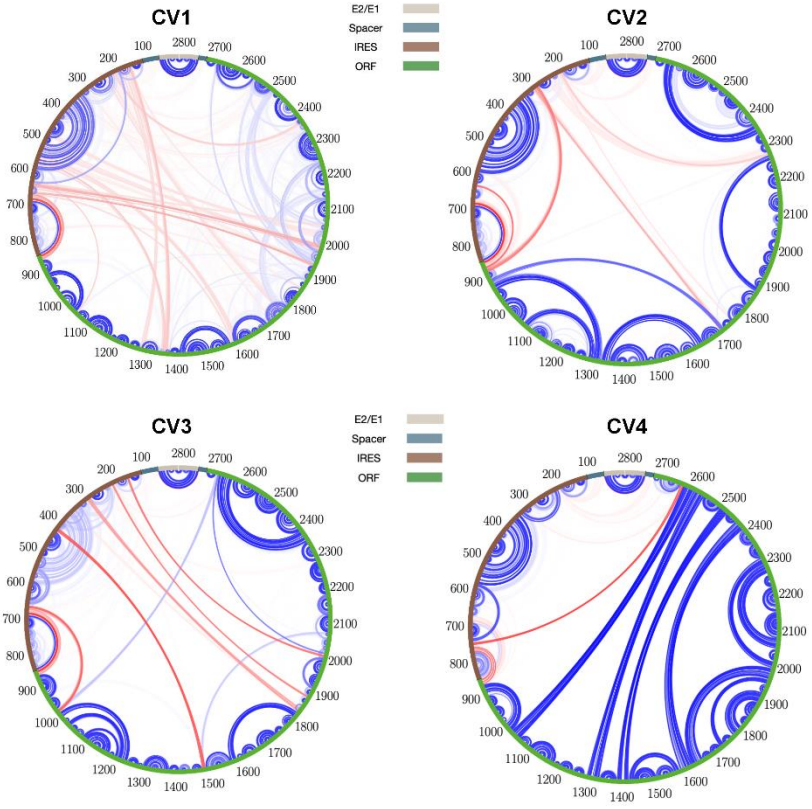
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Extended Data Figure 13



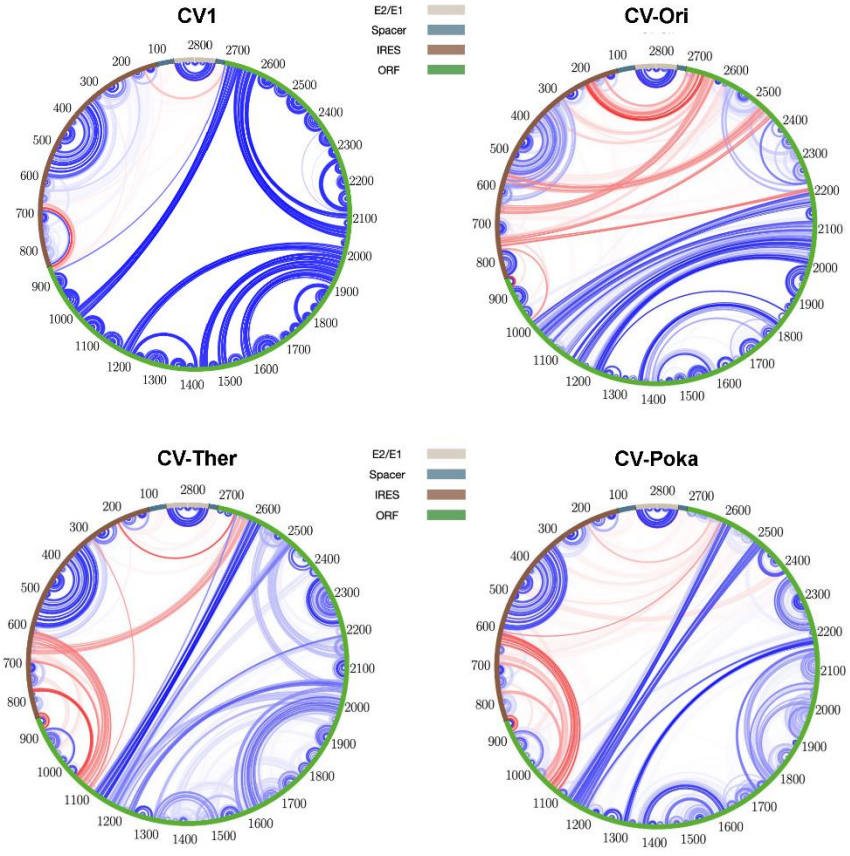
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Extended Data Figure 14



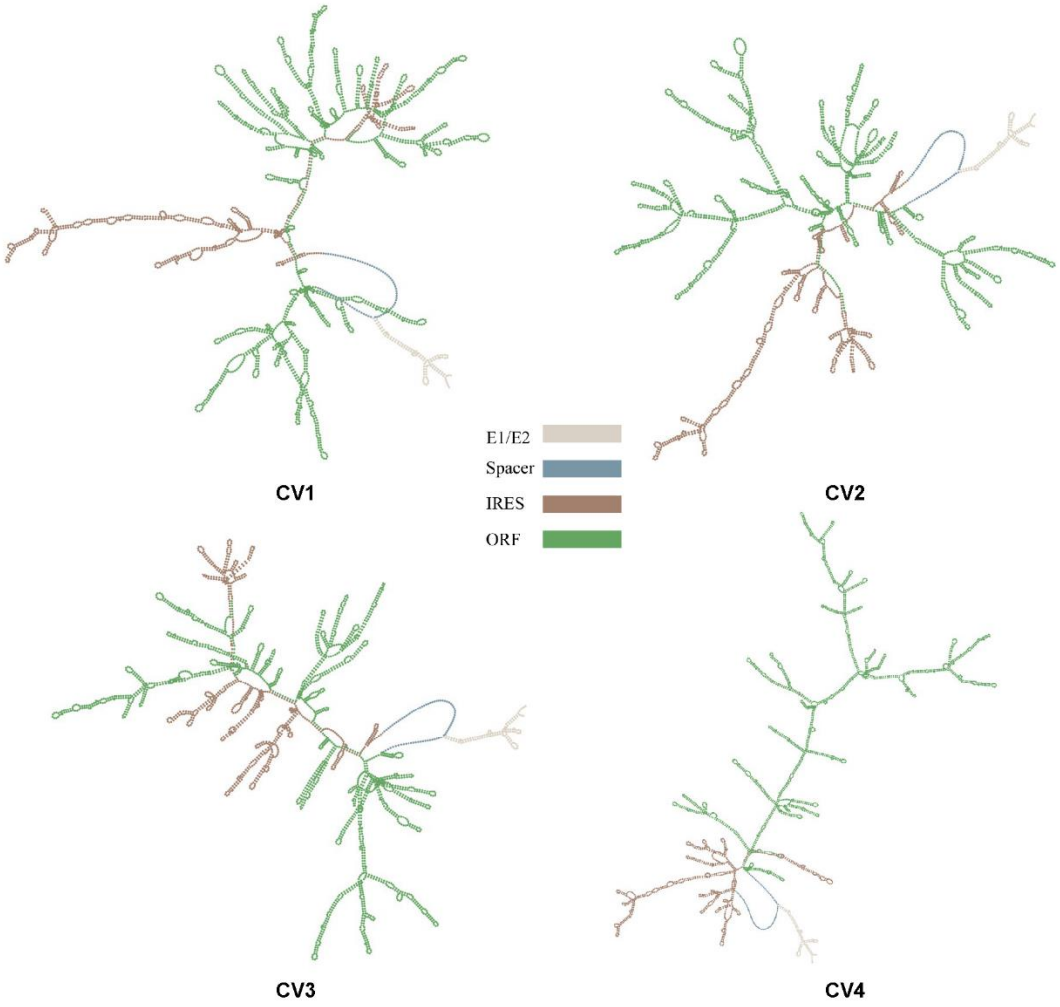
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Extended Data Figure 15



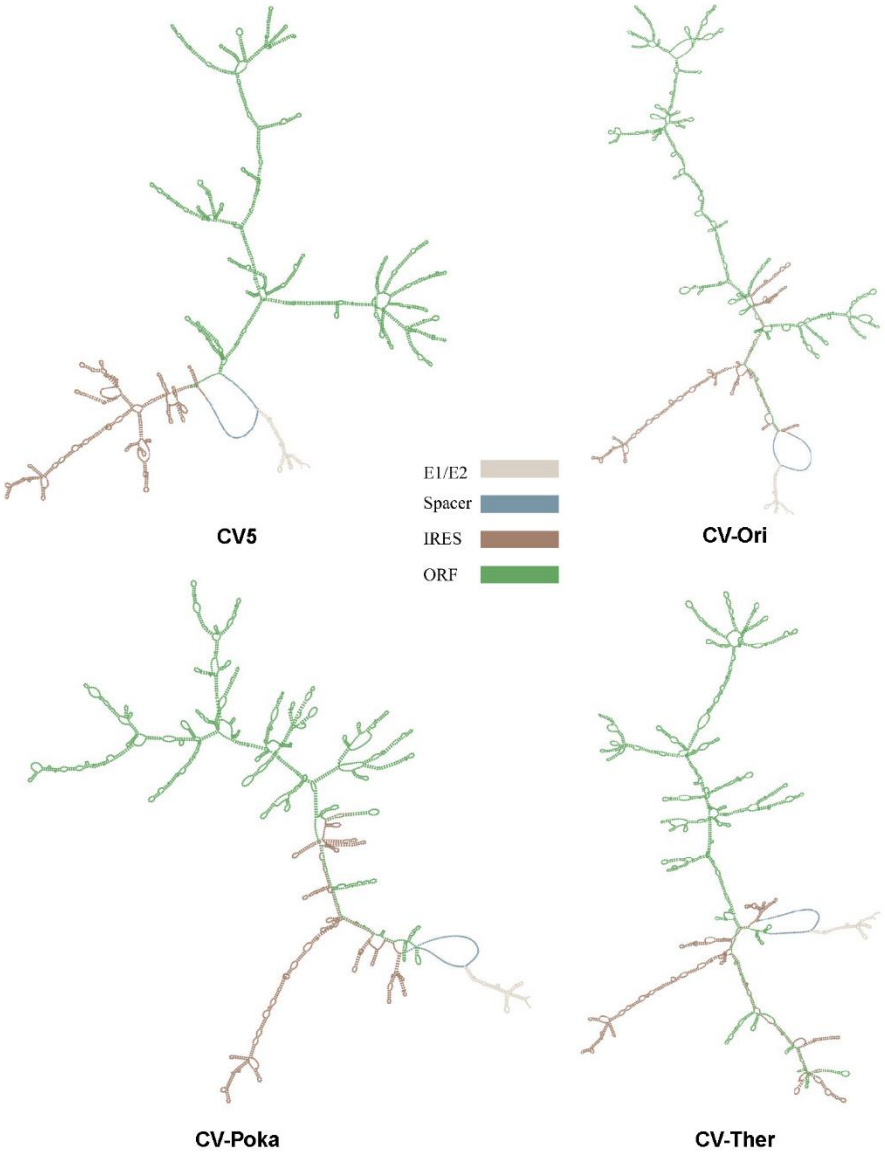
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Extended Data Figure 16



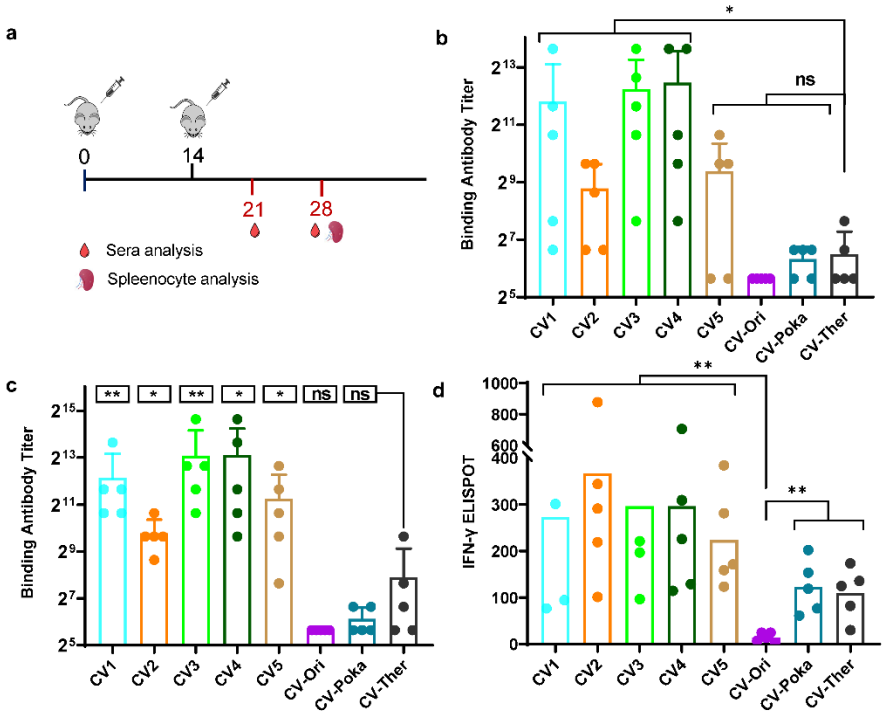
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Extended Data Figure 17



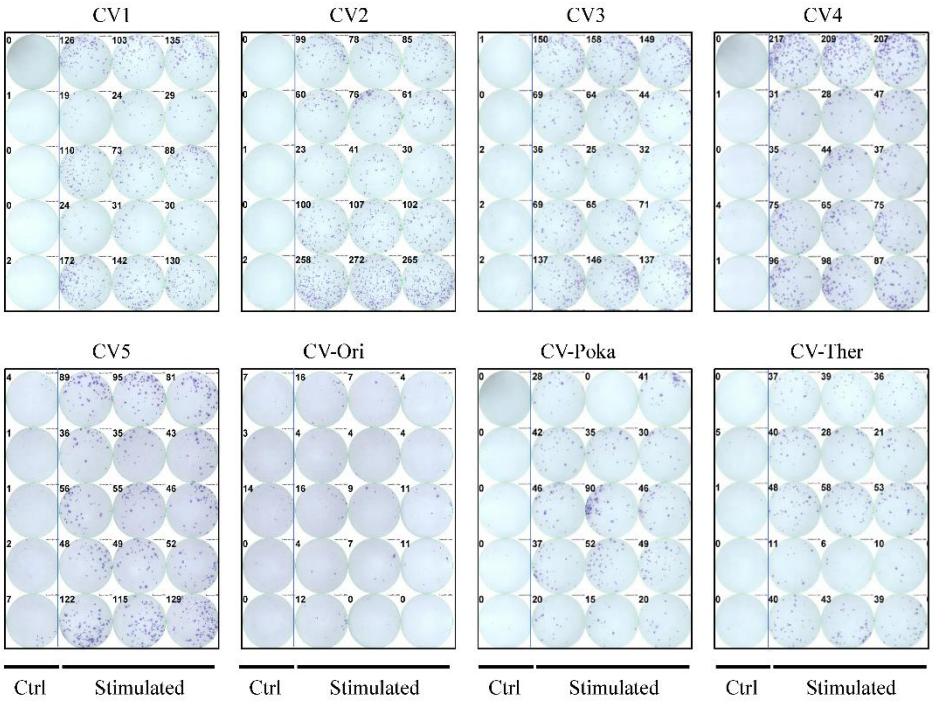
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Extended Data Figure 18



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Extended Data Figure 19



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Extended Data Table 1

Seq	MFE (kcal/mol)	CAI	Ensemble Diversity	Positional Entropy	T_m (°C)	Circularization efficiency (%)	In-solution $t_{1/2}$ (h)	Decay constant (10^{-2})	In-cell $t_{1/2}$ (h)	MFI	D28 pVNT	ELISPOT
CR1	-1338.5	0.81	169.11	354.8	79.41	81.53	20.62	3.36	24.34	21979	170511	169.8
CR2	-1301.3	0.81	217.9	483.0	78.08	77.00	16.84	4.12	19.31	22302.7	90007.6	243.1
CR3	-1298.2	0.87	161.46	335.3	78.11	77.93	15.56	4.46	24.55	22826	134644	280.1
CR4	-1219.7	0.87	164.12	337.6	73.92	80.80	12.44	5.57	15.46	22126	87136.5	253.3
CR5	-1223.6	0.93	179.96	377.0	76.18	70.97	12.05	5.75	9.18	22183	71741.8	210.4
CR6	-1142.3	0.93	213.26	447.4	73.38	68.40	11.37	6.10	18.32	17863.7	60288.4	101.2
CR-Ther	-900.6	0.93	609.31	1410.8	68.59	60.80	9.68	7.16	13.60	18295.7	66552.3	69.3
CR-Novo	-894.9	0.82	576.49	1259.2	69.00	61.53	9.14	7.58	10.21	21365	100432	113.4

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Extended Data Table 2

Tissue	6h		24h		48h		72h		168h	
	CR-Novo	CR3	CR-Novo	CR3	CR-Novo	CR3	CR-Novo	CR3	CR-Novo	CR3
Injection site	2.21×10^{10}	4.42×10^{10}	1.55×10^{8}	3.11×10^{9}	2.39×10^{7}	9.67×10^{7}	1.06×10^{7}	1.13×10^{8}	5.71×10^{6}	1.89×10^{6}
Lymph node	1.78×10^{7}	1.17×10^{8}	5.52×10^{6}	3.38×10^{8}	3.56×10^{7}	2.71×10^{7}	7.08×10^{7}	2.54×10^{8}	4.10×10^{7}	5.47×10^{7}
Spleen	1.58×10^{8}	1.24×10^{8}	4.17×10^{7}	1.63×10^{8}	3.27×10^{7}	1.11×10^{7}	3.30×10^{7}	1.06×10^{7}	1.22×10^{7}	1.28×10^{7}
Liver	1.90×10^{7}	2.72×10^{7}	2.80×10^{5}	7.35×10^{5}	3.26×10^{5}	9.18×10^{4}	1.98×10^{5}	6.37×10^{4}	1.22×10^{5}	9.60×10^{4}
Lung	7.72×10^{6}	7.84×10^{6}	5.42×10^{5}	1.43×10^{6}	6.71×10^{5}	8.89×10^{5}	1.11×10^{6}	5.96×10^{5}	1.83×10^{5}	1.47×10^{5}
Heart	2.08×10^{7}	5.97×10^{6}	9.14×10^{5}	2.59×10^{5}	3.04×10^{5}	8.53×10^{5}	1.08×10^{5}	4.19×10^{5}	2.03×10^{5}	2.56×10^{5}
Kidney	3.13×10^{6}	2.33×10^{6}	6.71×10^{4}	1.48×10^{5}	1.66×10^{5}	1.01×10^{6}	1.62×10^{5}	3.13×10^{6}	2.04×10^{5}	1.32×10^{5}
Brain	8.59×10^{5}	8.87×10^{5}	2.74×10^{6}	1.73×10^{6}	1.74×10^{5}	3.47×10^{5}	4.17×10^{5}	4.63×10^{5}	1.36×10^{5}	2.37×10^{5}
Sm. Int.	3.85×10^{5}	1.55×10^{6}	1.25×10^{6}	2.26×10^{5}	1.84×10^{5}	1.42×10^{5}	1.55×10^{5}	2.39×10^{5}	1.10×10^{5}	1.35×10^{5}

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Extended Data Table 3

Seq	MFE (kcal/mol)	CAI	Ensemble Diversity	Positional Entropy	T_m (°C)	Circularization efficiency (%)	In-solution $t_{1/2}$ (h)	MFI (24h)	D28 Ab GMT	ELISPOT
CV1	-862.80	0.93	525.21	1201.5	73.73	87.6	9.00	24515	4480	272.7
CV2	-899.90	0.92	308.43	647.5	76.20	79.6	13.62	26735	880	366.2
CV3	-979.50	0.90	351.68	728.0	76.85	82.4	13.82	28858	8640	295.8
CV4	-1129.20	0.85	243.61	499.9	79.85	89.3	18.71	27836	8800	296.2
CV5	-1193.80	0.79	228.75	495.3	81.98	87.4	122.2	34402	2440	223.6
CV-Ori	-535.50	0.67	624.23	1431.6	71.92	78.6	5.49	6462	50	13.3
CV-Poka	-679.90	1.00	755.43	1735.4	73.93	63.6	9.39	12204	70	122.4
CV-Ther	-634.00	0.94	676.78	1497.6	74.65	60.9	8.88	19042	240	109.1